Ruukkilabra
UPPER SECONDARY PHYSICS

FY1

Physics as a natural science

KIVIRUUKIN LUKIO

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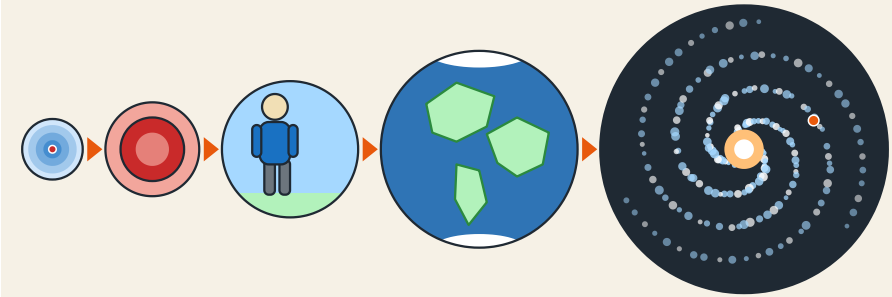
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FY1 · CHAPTER 1

Physics as a science and the scales of the universe

> **Why** can physics study both a proton and the whole universe, even though their sizes are so far apart?

Simulation, tasks and model solutions online:
`ruukkilab.fi/en/fy1-1.html`



IN THIS LESSON, YOU WILL LEARN TO

- describe the structure of matter from molecules to quarks and Earth's place in the structures of the universe.
- write a size in scientific notation and compare sizes using orders of magnitude.
- recognise the four fundamental interactions and link a phenomenon to each.

PRIOR KNOWLEDGE

- Powers from lower secondary school mathematics, for example $10^3 = 1000$ and $10^{-2} = 0.01$
- The units of length: metre, millimetre and kilometre

Core path: follow the steps in order. Besides calculations, the core path includes investigations, explanations and graphs. Tasks marked *Optional* give extra practice. Optional tasks: 6.

Think about it

Physics studies the smallest and the largest structures we know with the same principles. The size of a proton is about $1.7 \cdot 10^{-15}$ m and the diameter of the observable universe about $8.8 \cdot 10^{26}$ m. Neither can be seen with the naked eye or measured with a metre rule, and yet their sizes are known.

> **What is matter made of, what larger structures is Earth a part of, and how many powers of ten are there between the smallest and the largest size?**

Theory

Key idea

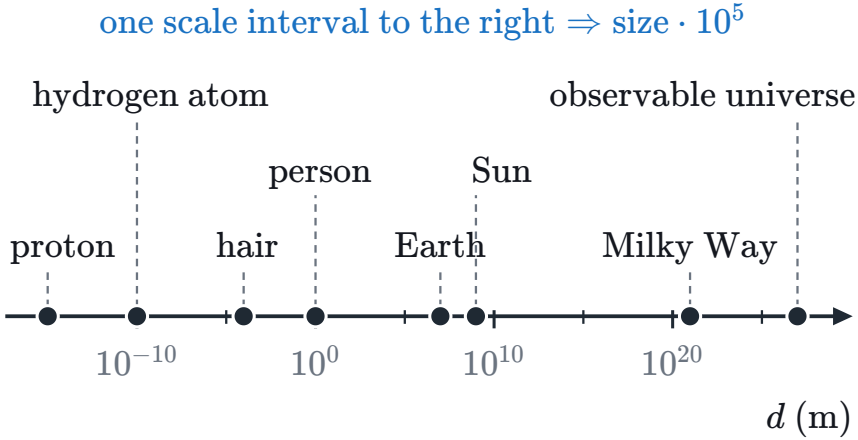
Physics builds models and tests them

is a natural science that looks for regularities in nature and explains them with models. A is a simplified description of how a phenomenon works. From a model you can work out a prediction, and a measurement can disprove that prediction. So physics does not rely on who says something, but on what experiments and measurements show.

Research proceeds in cycles. An observation raises a question, a model is proposed to answer it, and a prediction is derived from the model. The prediction is tested in an experiment, and the result either supports the model or forces us to change it. A measurement gives its result as a number and a unit: the SI unit of length is the metre, m.

Sizes vary by powers of ten

The objects of physics come in hugely different sizes: a proton is about $1.7 \cdot 10^{-15}\ \mathrm{m}$ and the observable universe about $8.8 \cdot 10^{26}\ \mathrm{m}$. A range of sizes this wide is awkward to present and compare in ordinary decimal form. That is why we use **scientific notation** $a \cdot 10^{n}$, where $1 \leq a < 10$ and n is an integer. When the exponent n grows by one, the number becomes ten times larger. The **Planck length**, about $1.6 \cdot 10^{-35}\ \mathrm{m}$, is the scale at which the effects of quantum gravity are expected to become significant. It has not been measured directly but calculated from constants of nature.



Order of magnitude tells the rough size

The is the power of ten that is closest to the number on the axis of powers of ten. The order of magnitude of $a \cdot 10^n$ is 10^n when $a < \sqrt{10} \approx 3.16$, and 10^{n+1} when $a > \sqrt{10}$. The limit $\sqrt{10}$ lies halfway between the points 10^n and 10^{n+1} on the axis of powers of ten. A person's height $1.7\ \mathrm{m}$ is of the order $10^0\ \mathrm{m}$, but the height of Mount Everest, $8.8 \cdot 10^3\ \mathrm{m}$, is of the order $10^4\ \mathrm{m}$.

The sizes of objects are rounded values: people, ants and galaxies are not all the same size. That is why the order of magnitude is often enough for a comparison, and it is worth asking how accurate the numbers are.

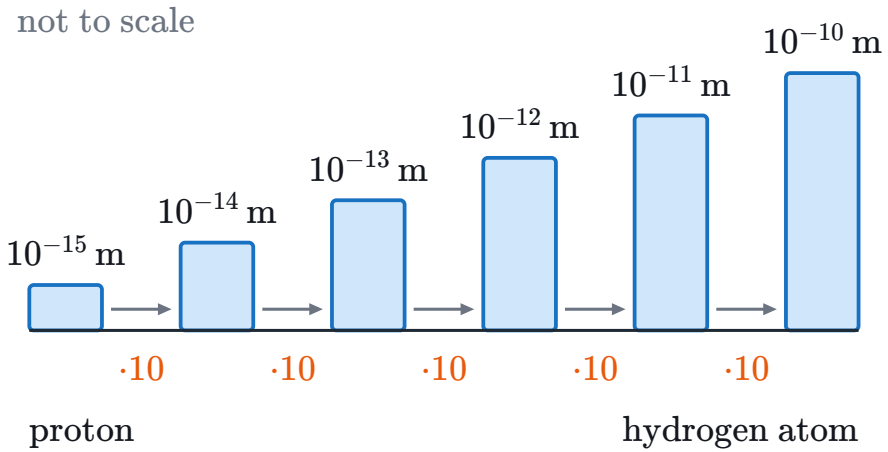
How many times larger?

When you want to know how many times larger one thing is than another, divide the larger size by the smaller:

$$k = \frac{d_{\text{larger}}}{d_{\text{smaller}}}$$

Here k is the ratio of the sizes, that is, how many times larger the object is. Both sizes must be in the same unit. Then the unit cancels, and k is just a number.

Powers of ten let you estimate the ratio in your head. Each step from one power of ten to the next is a tenfold size. A hydrogen atom is of the order $10^{-10}\ \text{m}$ and a proton $10^{-15}\ \text{m}$. There are five steps between them, so the atom is about $10 \cdot 10 \cdot 10 \cdot 10 \cdot 10 = 10^5$, that is, a hundred thousand times larger than the proton. The number of steps is the difference of the exponents: $-10 - (-15) = 5$.



$$5 \text{ steps} \Rightarrow 10^5 = 100\,000 \text{ times larger}$$

A light-year is a unit of length

Distances between stars are so large that the metre is impractical. A **light-year** is the distance that light travels in a vacuum in one year, about $9.46 \cdot 10^{15}\ \text{m}$. So it is a length, not a time. Light travel times and the

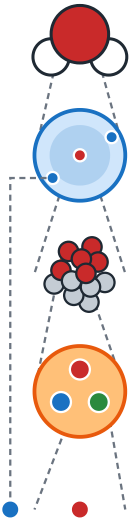
astronomical units of length are covered in more detail in the chapter *Quantities, units and the SI system*.

The structure of matter

Matter consists of nested structures

All the matter around us consists of **atoms**. Atoms can join together to form : a water molecule has one oxygen atom and two hydrogen atoms. At the centre of every atom there is a small , and **electrons** move around it. The nucleus consists of and usually also . A proton and a neutron each consist of three . Quarks and the electron are : as far as we know they have no smaller parts, and no size has been measured for them. According to experiments they are smaller than about $10^{-18}\,\mathrm{m}$.

not to scale



water molecule: three atoms together
$3 \cdot 10^{-10}\,\mathrm{m}$

atom: nucleus and electrons
$10^{-10}\,\mathrm{m}$

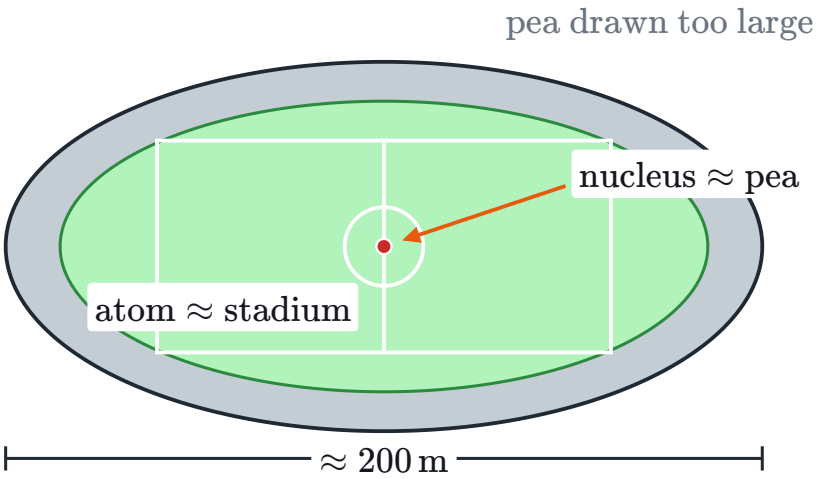
nucleus: protons and neutrons
10^{-15}–$10^{-14}\,\mathrm{m}$

proton or neutron: 3 quarks
$10^{-15}\,\mathrm{m}$

quark, electron: elementary particles
$< 10^{-18}\,\mathrm{m}$

An atom is tens of thousands of times larger than its nucleus

The diameter of an atom is about $10^{-10}\ \mathrm{m}$, but that of the nucleus is only 10^{-15}–$10^{-14}\ \mathrm{m}$. There are four or five orders of magnitude between them, so the atom is tens of thousands of times larger than its nucleus. If an atom were enlarged to the size of a football stadium, the nucleus at its centre would be the size of a pea. Even so, the nucleus holds over 99.9 percent of the mass of the atom. The size of the atom is set by the region where the electrons move, and the nucleus is small and very dense at its centre. The nucleus of a hydrogen atom is a single proton.



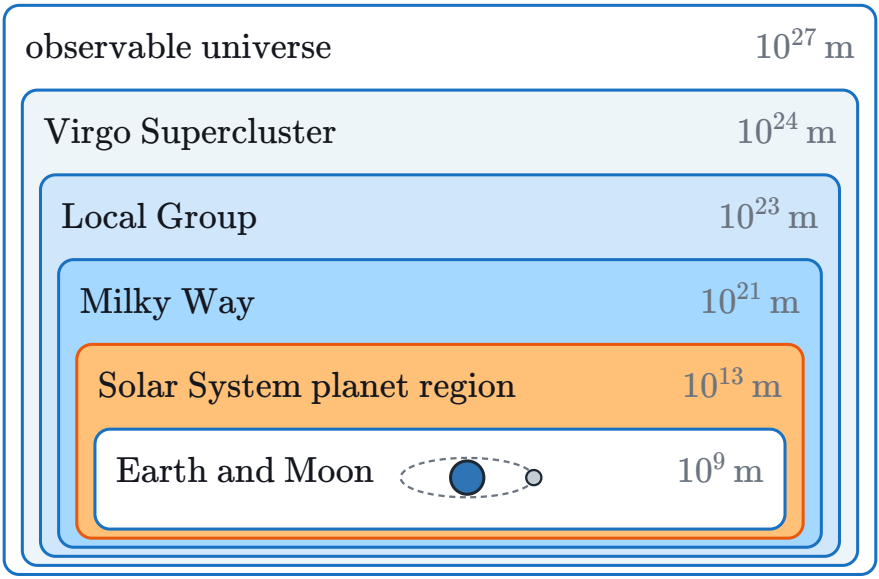
Structures of the universe

Earth's address in the universe

Earth is a **planet**: it orbits a **star**, the Sun, and the Moon orbits Earth. The energy of a Sun-like star is released in fusion reactions in its core, and the star radiates energy into space. In visible light, the planets of the Solar System are seen because of the sunlight they reflect. They also emit their own thermal radiation. The Sun, its eight planets with their moons, and

the dwarf planets, asteroids and comets form the . It is one **planetary system** among many: thousands of planets orbiting other stars are already known.

The Sun is one of the hundreds of billions of stars in the Milky Way. The **Milky Way** is a , that is, a huge collection of stars, gas and dust held together by gravity, and its diameter is about 10^5 light-years. There are an estimated hundreds of billions of galaxies in the observable universe. They form **groups of galaxies** and larger **clusters of galaxies**: the Milky Way and the Andromeda Galaxy belong to the Local Group. Groups and clusters form superclusters, and on the largest scale the galaxies lie in web-like filaments with almost empty regions between them.



The edge of the Solar System depends on what is measured

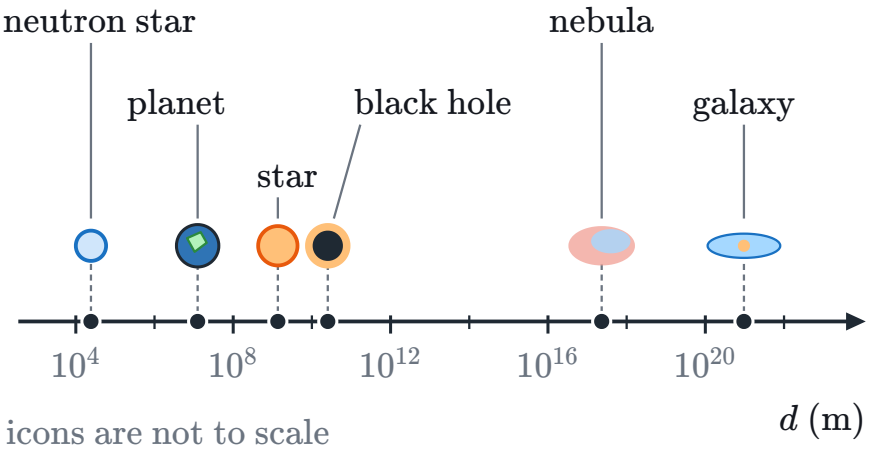
The Solar System has no single sharp edge. Particles flow out from the Sun, the **solar wind**, which blows a bubble around it, the **heliosphere**. Its boundary, the heliopause, is about $120\ \mathrm{AU}$ from the Sun, and the

Voyager probes have crossed it. However, the Sun's gravity dominates much farther away. From the orbits of long-period comets we conclude that a swarm of comets, the Oort cloud, orbits the Sun far away, about $2000–100\,000\ \mathrm{AU}$ from it. So the Voyagers are in interstellar space but still within the reach of the Sun's gravity.

Types of objects, not steps

Not all astronomical objects are steps in the same nested series. Instead, they are found inside galaxies, among the stars. A **nebula** is a cloud of gas and dust; for example, new stars are forming in the Orion Nebula. A is the remnant of the explosion of a massive star. Its mass is usually greater than that of the Sun, but its diameter is only about $20–25\ \mathrm{km}$, because the matter has been squeezed into almost pure neutrons.

A is a region from which, because of gravity, nothing, not even light, can escape. It has no surface: its size is given as the diameter of its boundary, the **event horizon**, which grows with the mass. The black hole at the centre of the Milky Way has a mass of about four million solar masses and an event horizon diameter of about $2.5 \cdot 10^{10}\ \mathrm{m}$. So size does not tell the type of object: a neutron star is smaller than Earth, and this black hole is larger than the Sun.



The observable universe is not the whole universe

The **observable universe** is the part of the universe from which light has had time to reach us during the age of the universe, about 13.8 billion years. Its diameter is about $8.8 \cdot 10^{26}$ m, or about 93 billion light-years. The number is greater than twice 13.8 billion light-years because space has expanded during the light's journey. The boundary of the observable universe is not the edge of the universe but the limit of our horizon: for an observer in another galaxy, the observable universe is a sphere around that observer. The size of the whole universe is not known, and it may be infinite.



The first deep field image of the James Webb Space Telescope, 2022. Almost every point of light is a distant galaxy.

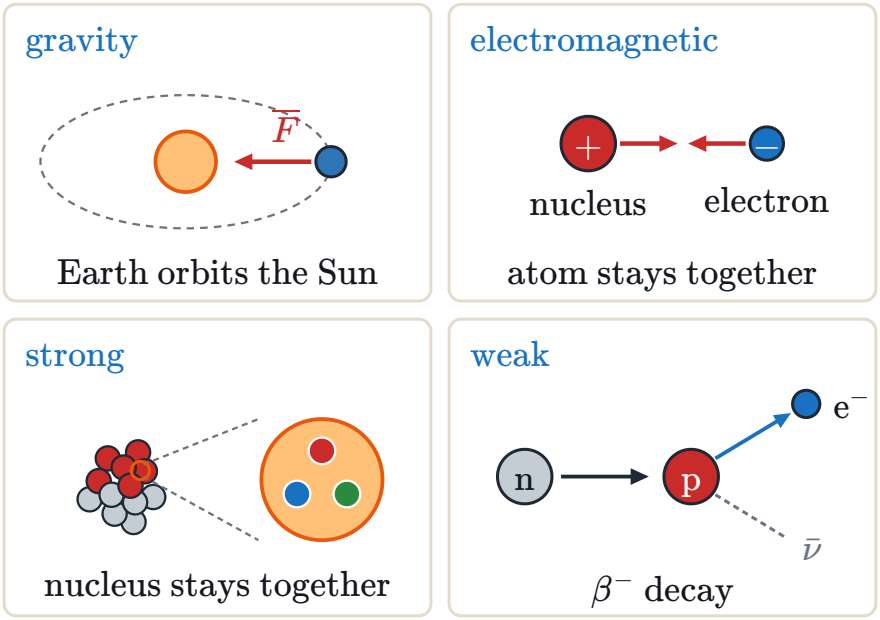
Photo: NASA, ESA, CSA, STScI, public domain

Four fundamental interactions

Structures stay together and change because of **interactions**. All known interactions come down to four **fundamental interactions**:

- **Gravity** acts between all objects that have mass, and it is always attractive. It keeps the Moon around Earth and the planets in their orbits around their stars, and it gathers clouds of gas into stars and galaxies.
- **The electromagnetic interaction** acts between electrically charged particles: opposite charges attract and like charges repel each other. It keeps the electrons around the nucleus in an atom, binds atoms into molecules and causes electric and magnetic phenomena, such as the ability of a rubbed balloon to attract bits of paper.

- **The strong interaction** binds quarks into protons and neutrons. Its residual effect holds protons and neutrons together in a nucleus, even though the protons repel each other electrically. It reaches only a distance the size of a nucleus, about $10^{-15}\ \mathrm{m}$.
- **The weak interaction** causes some radioactive decays. For example, in **beta-minus decay** (β^- decay) a neutron in the nucleus changes into a proton, and an electron and a particle called an antineutrino leave the nucleus. The weak interaction is also involved in the Sun's energy production.



Scale decides which interaction dominates

The same four interactions act everywhere, from inside the nucleus to clusters of galaxies, but the scale decides which of them dominates. The strong and weak interactions reach only the scale of the nucleus. Between particles the electromagnetic interaction is enormously stronger than gravity, but large objects are almost neutral: their positive

and negative charges cancel each other's effect. Between particles gravity is the weakest, but it is always attractive and grows with mass. That is why gravity dominates on the scale of planets, stars and galaxies.

Common misconception

A difference in powers of ten is not a difference in sizes. If the orders of magnitude of two objects differ by six, the ratio of their sizes based on the orders of magnitude is about 10^6, that is, a million, not six. In addition, objects of very different sizes cannot all be recognised in the same scale picture if they are drawn in the right proportions of size.

Neutron stars and black holes are not steps larger than a galaxy, but objects inside galaxies. A black hole also does not suck in everything around it: far from it, its gravity is the same as that of any object of equal mass, and stars can orbit it like planets orbit the Sun.

Summary

- Physics builds models from which predictions are derived, and an experiment or measurement decides whether the model holds.
- Scientific notation $a \cdot 10^n$ writes very small and very large numbers briefly; one step in the exponent means a tenfold size.
- The order of magnitude is the nearest power of ten on the axis of powers of ten. The ratio of sizes is found by dividing the larger size by the smaller, and each step of a power of ten is a tenfold size.
- A light-year is the distance light travels in a year, about $9.46 \cdot 10^{15}\ \mathrm{m}$.
- Molecules consist of atoms, atoms of a nucleus and electrons, the nucleus of protons and usually also neutrons. Protons and neutrons consist of quarks. An atom is tens of thousands of times larger than its nucleus.

- Earth is a planet that orbits a star. The Solar System is one planetary system in the Milky Way, and the Milky Way is one of the hundreds of billions of galaxies of the observable universe.
- Nebulae, neutron stars and black holes are types of objects inside galaxies, not steps of structure.
- The four fundamental interactions are gravity and the electromagnetic, strong and weak interactions. On small scales the interactions of the nucleus and the atom dominate, and on large scales gravity.

Examples

Once you have studied the chapter's theory, move on to the guided examples. In the first you follow a worked solution step by step; in the next ones you make the choices yourself, and the hints help if you get stuck.

Example 1: Hydrogen atom and proton

The diameter of a hydrogen atom is about $1.1 \cdot 10^{-10}\ \mathrm{m}$ and the diameter of a proton about $1.7 \cdot 10^{-15}\ \mathrm{m}$. How many times larger is a hydrogen atom than a proton, and how many powers of ten are there between their orders of magnitude?

STEP 1/7

Given and find. The diameter of the hydrogen atom is $d_{\text{atom}} = 1.1 \cdot 10^{-10}\ \mathrm{m}$ and the diameter of the proton is $d_{\text{proton}} = 1.7 \cdot 10^{-15}\ \mathrm{m}$. We are asked for the ratio of the sizes k and the number of steps of powers of ten.

STEP 2/7

Principle. The question "how many times larger" is a division: divide the larger size by the smaller. The number of steps is the difference of the exponents of the orders of magnitude.

STEP 3/7

Rearranged formula.

$$k = \frac{d_{\text{atom}}}{d_{\text{proton}}}$$

STEP 4/7

Unit conversions. None needed: both sizes are in metres, so the unit cancels and k is just a number.

STEP 5/7

Substitution.

$$\begin{aligned} k &= \frac{d_{\text{atom}}}{d_{\text{proton}}} \\ &= \frac{1.1 \cdot 10^{-10}\,\text{m}}{1.7 \cdot 10^{-15}\,\text{m}} \\ &= 6.47\ldots \cdot 10^{4} \\ &\approx 6.5 \cdot 10^{4} \end{aligned}$$

On a calculator, the division $1.1 \cdot 10^{-10} : 1.7 \cdot 10^{-15}$ is done by entering the powers of ten with the power-of-ten key, which is often called EE or EXP.

STEP 6/7

Result. A hydrogen atom is about $6.5 \cdot 10^{4}$, that is, about 65 000 times larger than a proton. The result is given with two significant figures like the initial data. The orders of magnitude are $10^{-10}\ \mathrm{m}$ and $10^{-15}\ \mathrm{m}$, so the number of steps is $-10-(-15)=5$.

STEP 7/7

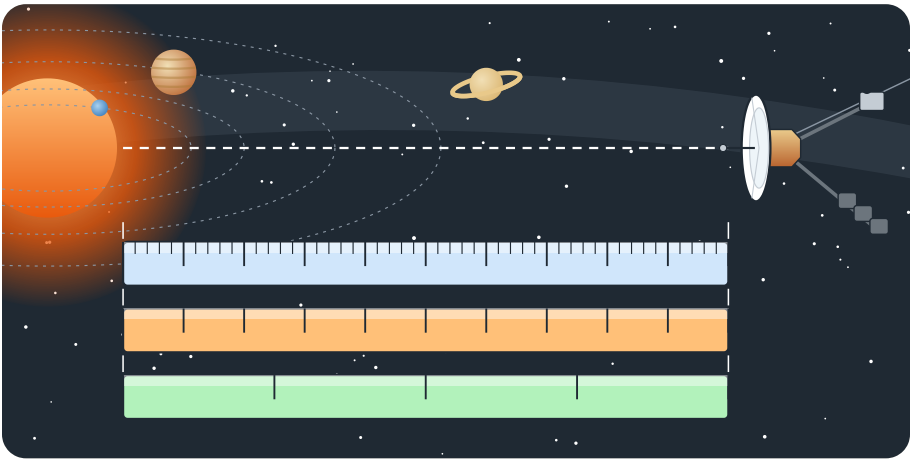
Reasonableness. Five steps mean about 10^{5}, that is, a hundred-thousandfold size, and $6.5 \cdot 10^{4}$ is close to that. Order-of-magnitude check: if the proton were enlarged to the size of a millimetre, the atom would be $65\,000\ \mathrm{mm} = 65\ \mathrm{m}$, that is, of the order of tens of metres.

FY1 · CHAPTER 2

Quantities, units and the SI system

Why is the distance to the Voyager 2 probe sometimes given in kilometres, sometimes in astronomical units and sometimes in light-hours, even though it is one and the same distance?

Simulation, tasks and model solutions online:
`ruukkilab.fi/en/fy1-2.html`



IN THIS LESSON, YOU WILL LEARN TO

- convert units using prefixes and powers of ten.
- give the result of a calculation with the right number of significant figures.
- give the same distance in kilometres, in astronomical units and as the travel time of light.

PRIOR KNOWLEDGE

- Sizes range from the Planck length to the observable universe over more than sixty powers of ten
  chapter *Physics as a science and the scales of the universe*
- Each step of a power of ten makes a length ten times greater or one tenth of what it was
  chapter *Physics as a science and the scales of the universe*

Core path: follow the steps in order. Besides calculations, the core path includes investigations, explanations and graphs. Tasks marked *Optional* give extra practice. Optional tasks: 5.

Think about it

NASA's Voyager 2 probe was launched in 1977, and it has crossed the heliopause, the edge of the bubble blown by the solar wind. It is now travelling in interstellar space, although it is still within the Sun's gravitational

influence. We still keep in contact with it by radio signals. The distance of the probe is sometimes given in kilometres, sometimes in astronomical units and sometimes as the time a signal takes to travel from there to here.

> **Why is the same distance given in different units, and what must be done between the numbers so that they mean the same thing?**

Theory

Key idea

Quantity, numerical value and unit

Physics describes the world with measurable properties called : length, mass, time and speed are quantities. The value of a quantity is measured by comparing it with an agreed measure, a , and the value is written as the product of a numerical value and a unit, for example the length of a pencil $l = 12.0\,\mathrm{cm}$. A numerical value without a unit tells you nothing: the number 12 could mean centimetres, hours or kilograms. The symbol of a quantity is written in italics (l, m, t) and the unit in upright letters ($\mathrm{m}, \mathrm{kg}, \mathrm{s}$).



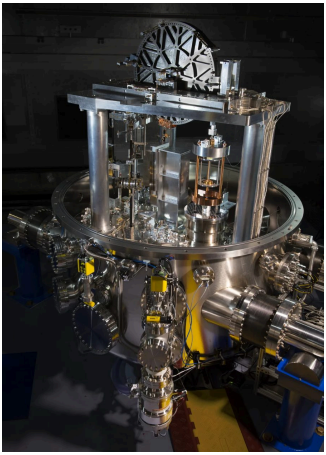
$$\underbrace{l}_{\text{quantity}} = \overbrace{\underbrace{12.0}_{\text{numerical value}} \quad \underbrace{\text{cm}}_{\text{unit}}}^{\text{value of the quantity}}$$

The SI system

So that a metre is the same metre everywhere in the world, the scientific community has agreed on a common system of units, the . It has seven base quantities, and each has its own base unit. All other quantities are derived from them: speed is length divided by time, so its unit is $\mathrm{m/s}$, and the unit of area is m^2.

Base quantity	Symbol	SI unit
length	l	metre m
mass	m	kilogram kg
time	t	second s
electric current	I	ampere A
temperature	T	kelvin K
amount of substance	n	mole mol
luminous intensity	I_{v}	candela cd

In this course you need the first three most. You will meet the other base quantities in later courses.

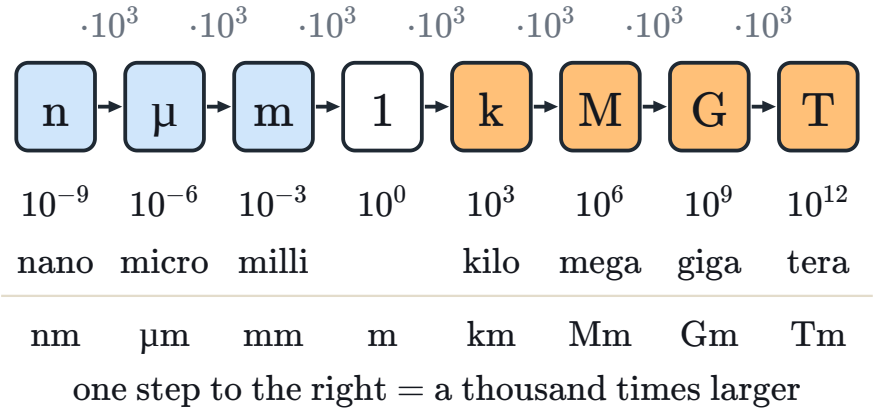


The NIST-4 Kibble balance at the US National Institute of Standards and Technology. Today the kilogram is defined using a constant of nature, the Planck constant, and a balance like this can measure mass according to that definition.

Photo: Jennifer Lauren Lee / NIST, public domain (PD-USGov-NIST)

Prefixes and powers of ten

The metre is good for measuring a room but not an atom or Neptune. That is why a , an abbreviation for a power of ten, is attached in front of the unit: $1\,\mathrm{km} = 10^3\,\mathrm{m}$ and $1\,\mathrm{mm} = 10^{-3}\,\mathrm{m}$. The most common prefixes go up in steps of a factor of a thousand (exceptions are, for example, centi 10^{-2} and deci 10^{-1}), and large and small numbers are written in scientific notation $a \cdot 10^n$, where $1 \leq a < 10$ and n is an integer. For example, the distance between the Earth and the Sun is $1.5 \cdot 10^{11}\,\mathrm{m} = 1.5 \cdot 10^{8}\,\mathrm{km}$.



When you move to a smaller unit, the numerical value grows: $4.2\,\mathrm{km} = 4.2 \cdot 10^3\,\mathrm{m} = 4200\,\mathrm{m}$. When you move to a larger unit, the numerical value gets smaller. The quantity stays the same, only the way of writing it changes.

A unit for every scale

A unit is chosen so that the numerical value is easy to read. In the chapter *Physics as a science and the scales of the universe* you saw that the structures of matter and of the universe lie on steps of powers of ten. Each step has a convenient unit:

- , about $10^{-15}\,\mathrm{m}$: femtometre, $1\,\mathrm{fm} = 10^{-15}\,\mathrm{m}$.
- , about $10^{-10}\,\mathrm{m}$: nanometre, $1\,\mathrm{nm} = 10^{-9}\,\mathrm{m}$.
- **Red blood cell**, about $10^{-5}\,\mathrm{m}$: micrometre, $1\,\mu\mathrm{m} = 10^{-6}\,\mathrm{m}$.
- **Earth**, about $10^{7}\,\mathrm{m}$: kilometre, $1\,\mathrm{km} = 10^{3}\,\mathrm{m}$.
- **Orbits of the planets**, about $10^{12}\,\mathrm{m}$: astronomical unit, $1\,\mathrm{AU} \approx 1.5 \cdot 10^{11}\,\mathrm{m}$.
- **Distances between stars**, about $10^{16}\,\mathrm{m}$ and more: light-year, about $9.46 \cdot 10^{15}\,\mathrm{m}$.

For example, the diameter of a hydrogen atom is $0.11\ \mathrm{nm}$ and that of a proton $1.7\ \mathrm{fm}$. A unit does not change the size, only the way of writing it. The astronomical unit and the light-year are not SI units, but they are defined in metres, so you can always convert them to metres.

Significant figures

The of a numerical value tell how precisely the quantity is known. Starting from the first digit that is not zero, all digits, including the trailing zeros of a decimal number, are significant: $0.0250\ \mathrm{m}$ has three significant figures and $3.0\ \mathrm{m}$ has two. Scientific notation makes the precision visible: $3.00 \cdot 10^{8}$ has three significant figures and $3 \cdot 10^{8}$ has one. The trailing zeros of a whole number are unclear: from the Moon's mean distance $384\,400\ \mathrm{km}$ you cannot see whether the precision is a hundred, ten or one kilometre. Show the precision in scientific notation, for example $3.844 \cdot 10^{5}\ \mathrm{km}$ (four significant figures). In the tasks of this book a value given as a whole number ending in zeros, such as $10\ \mathrm{min}$, is exact unless stated otherwise.

The result of a multiplication or division is given with as many significant figures as the least precise factor has. Rounding is done only at the end, and exact numbers such as the conversion factor 1000 have no limiting effect. For example, $1.85\ \mathrm{m} \cdot 0.92\ \mathrm{m} = 1.702\ \mathrm{m}^2$ is rounded to $1.7\ \mathrm{m}^2$.

Unit conversions

When you change a unit, the quantity does not change, only the way of writing it. A conversion is done by replacing the old unit with its value in the new unit. Since $1\ \mathrm{km} = 1000\ \mathrm{m}$, you can write $1000\ \mathrm{m}$ in place of the kilometre:

$$4.2\ \mathrm{km} = 4.2 \cdot 1000\ \mathrm{m} = 4200\ \mathrm{m}$$

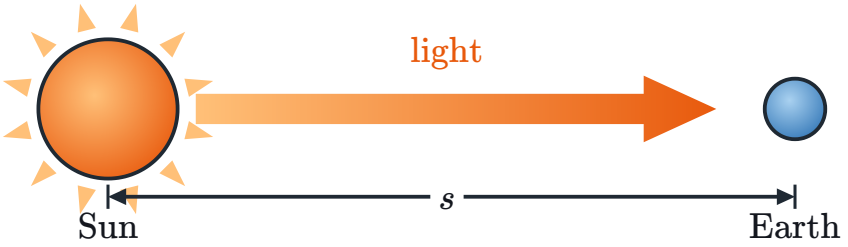
In the other direction $1\ \mathrm{m} = 0.001\ \mathrm{km}$, so $350\ \mathrm{m} = 350 \cdot 0.001\ \mathrm{km} = 0.35\ \mathrm{km}$. Always check the result by reasoning: in a smaller unit the numerical value is greater, because more of the small units are needed. For area, the unit is a product of two lengths, so $1\ \mathrm{km}^2 = 1000\ \mathrm{m} \cdot 1000\ \mathrm{m} = 10^6\ \mathrm{m}^2$.

In astronomy the kilometre is too small a unit. The **astronomical unit** AU is an agreed unit of length that is close to the mean distance between the Earth and the Sun: $1\ \mathrm{AU} = 149\,597\,870.7\ \mathrm{km} \approx 1.496 \cdot 10^8\ \mathrm{km}$. The light-year is also a unit of length, not of time: it is the distance light travels in a vacuum in a year, about $9.46 \cdot 10^{12}\ \mathrm{km}$.

A distance can also be given as the time light needs to travel it. Light travels in a vacuum at a constant speed c, so the distance it covers is speed times time, $s = ct$. The travel time is obtained by solving this formula for time:

$$t = \frac{s}{c}$$

where t is the travel time of light, s the distance travelled by light and $c = 299\,792\,458\ \mathrm{m/s} \approx 2.998 \cdot 10^5\ \mathrm{km/s}$ the speed of light in a vacuum. Light travels one AU in $\frac{1.496 \cdot 10^8\ \mathrm{km}}{2.998 \cdot 10^5\ \mathrm{km/s}} = 499\ \mathrm{s}$, which is $8.32\ \mathrm{min}$.



$$s = 1.496 \cdot 10^{8}\,\mathrm{km}$$

$$= 1\,\mathrm{AU}$$

$$\text{light travel time } t = 499\,\mathrm{s} = 8.32\,\mathrm{min}$$

Common misconception

The numerical value and the unit belong together. When the unit changes, the numerical value changes inversely: a larger unit means a smaller numerical value, and the quantity itself stays the same.

Square units are easy to forget. One square kilometre is not a thousand but a million square metres, because $(10^3)^2 = 10^6$.

The light-year is not a unit of time, even though the name contains the word year. It is a unit of length, about $9.46 \cdot 10^{12}\,\mathrm{km}$.

Summary

- The value of a quantity is a numerical value times a unit: $l = 12.0\,\mathrm{cm}$. Without a unit the number tells you nothing.
- In the SI system the base units are, for example, the metre, the kilogram and the second, and the other units are derived from them.

- A prefix is a power of ten: kilo 10^{3}, mega 10^{6}, giga 10^{9}, milli 10^{-3}, micro 10^{-6}.
- In a unit conversion, the old unit is replaced by its value in the new unit, for example $4.2\,\mathrm{km} = 4.2 \cdot 1000\,\mathrm{m}$. For a square unit the conversion is done twice: $1\,\mathrm{km}^2 = 10^{6}\,\mathrm{m}^2$.
- The result of a multiplication or division is given with as many significant figures as the least precise factor has.
- The travel time of light is $t = \frac{s}{c}$. Light travels one AU in about $8.3\,\mathrm{min}$.

Examples

Once you have studied the chapter's theory, move on to the guided examples. In the first you follow a worked solution step by step; in the next ones you make the choices yourself, and the hints help if you get stuck.

Example 1: Light from the Sun to Neptune

Neptune's mean distance from the Sun is $30.1\,\mathrm{AU}$. How many kilometres is that in scientific notation, and how many hours does light take to travel from the Sun to Neptune? Use the values $1\,\mathrm{AU} = 1.496 \cdot 10^{8}\,\mathrm{km}$ and $c = 2.998 \cdot 10^{5}\,\mathrm{km/s}$.

STEP 1/7

Given and unknown. The distance is $s = 30.1\,\mathrm{AU}$, $1\,\mathrm{AU} = 1.496 \cdot 10^{8}\,\mathrm{km}$ and the speed of light is $c = 2.998 \cdot 10^{5}\,\mathrm{km/s}$. We want the distance s in kilometres and the travel time t in hours.

STEP 2/7

Principle. The distance does not change when the unit changes, so in place of AU we write its value in kilometres. Light travels at a constant speed c, so the given formula $s = ct$ gives the travel time.

STEP 3/7

Rearranged formula.

$$t = \frac{s}{c}$$

STEP 4/7

Unit conversions. The distance is converted to kilometres by writing $1.496 \cdot 10^8\,\mathrm{km}$ in place of AU:

$$\begin{aligned} s &= 30.1\,\mathrm{AU} \\ &= 30.1 \cdot 1.496 \cdot 10^8\,\mathrm{km} \\ &= 4.503 \cdot 10^9\,\mathrm{km} \end{aligned}$$

The travel time comes out in seconds, because km cancels. The time is asked in hours, so the seconds are converted to hours: $1\,\mathrm{h} = 3600\,\mathrm{s}$.

STEP 5/7

Substitution.

$$\begin{aligned} t &= \frac{s}{c} \\ &= \frac{4.503 \cdot 10^{9}\,\mathrm{km}}{2.998 \cdot 10^{5}\,\mathrm{km/s}} \\ &= 1.502 \cdot 10^{4}\,\mathrm{s} \\ &= (1.502 \cdot 10^{4} : 3600)\,\mathrm{h} = 4.17\,\mathrm{h} \end{aligned}$$

STEP 6/7

Result. The distance is $4.50 \cdot 10^{9}\ \mathrm{km}$ and the travel time of light is $4.17\ \mathrm{h}$, to three significant figures like the initial value $30.1\ \mathrm{AU}$.

STEP 7/7

Reasonableness. Light travels one AU in $8.3\ \mathrm{min}$, so $30\ \mathrm{AU}$ takes light about

$$\begin{aligned} 30 \cdot 8.3\,\mathrm{min} &\approx 250\,\mathrm{min} \\ &\approx 4.2\,\mathrm{h} \end{aligned}$$

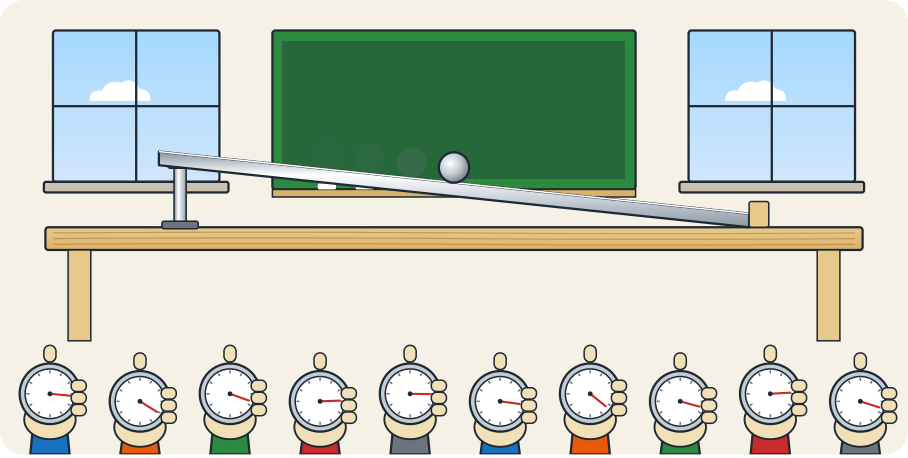
This agrees with the result, and the distance is of the order of billions of kilometres, as suits an outer planet.

FY1 · CHAPTER 3

Measuring and the reliability of a measurement

Why do ten students get ten different results for the rolling time of the same ball, even though they all use the same stopwatch?

Simulation, tasks and model solutions online:
`ruukkilab.fi/en/fy1-3.html`



IN THIS LESSON, YOU WILL LEARN TO

- tell gross error, random error and systematic error apart and recognise their sources.
- calculate a mean and estimate the scatter of the results using half of the range.
- correct a systematic error and judge whether a measurement is repeatable and accurate.

PRIOR KNOWLEDGE

- A measurement result is given as a numerical value and a unit, and significant figures tell how precise it is
  chapter *Quantities, units and the SI system*

Core path: follow the steps in order. Besides calculations, the core path includes investigations, explanations and graphs. Tasks marked *Optional* give extra practice. Optional tasks: 2.

Think about it

In a physics lesson the class measures the time a ball takes to roll from the top of a ramp to the bottom. Everyone has a similar hand stopwatch and presses it with their own hand. Still, the results are 2.18 s, 2.45 s, 2.68 s, 2.67 s, 2.40 s and so on. The teacher says that the true time of the ball exists, but does not tell it.

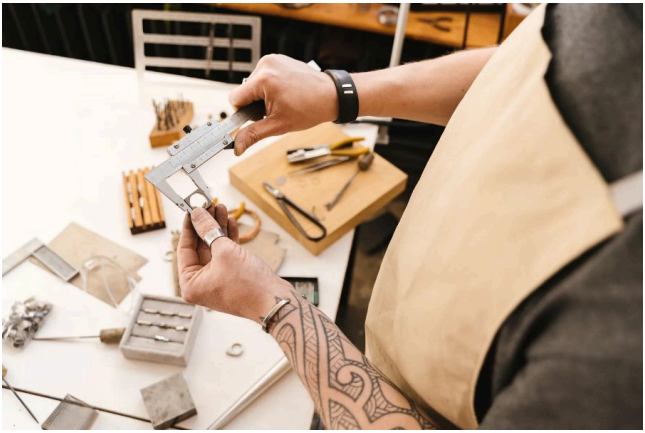
> **How can the class get a reliable result from these, and how can they know that the stopwatch is not wrong in the same way for everyone?**

Theory

Key idea

A **measurement** means determining the value of a quantity by comparing it with a unit. A measurement result is a numerical value and a unit, for example $2.43\ \mathrm{s}$. The result usually differs a little from the true value of the quantity, because reading the instrument, the method of measuring and the conditions all add a deviation to it. The difference $x - x_{\mathrm{true}}$ between the result and the true value is the . Every measurement result has some uncertainty, even one made by a skilled measurer. The true value is usually not known, so neither is the measurement error or its size. Therefore the **measurement uncertainty** is stated together with the result. It is an estimate of how large the measurement error can be. It does not usually mean an absolute largest possible deviation from the true value.

Errors are divided into three kinds: gross error, random error and systematic error. You can tell them apart by how they behave when the measurement is repeated.



With a vernier caliper a length can be read to a fraction of a millimetre. The precision of the measuring instrument is one source of uncertainty in a measurement result.

Photo: Antoni Shkraba, Pexels License

Gross error

A is a single clear mistake: the instrument is read wrongly, a number is recorded wrongly (for example $3.52\ \mathrm{s}$ instead of $3.25\ \mathrm{s}$), the stopwatch is accidentally started only once the ball is already moving, or the result is recorded in the wrong unit. It usually shows up as a result that differs clearly from the others. A gross error is not estimated or averaged but looked for: the odd result is checked, and the measurement can be repeated if necessary. A result is not removed merely because it differs from the others: it is left out of the calculations only if the cause of the error is found, and leaving it out is justified and recorded.

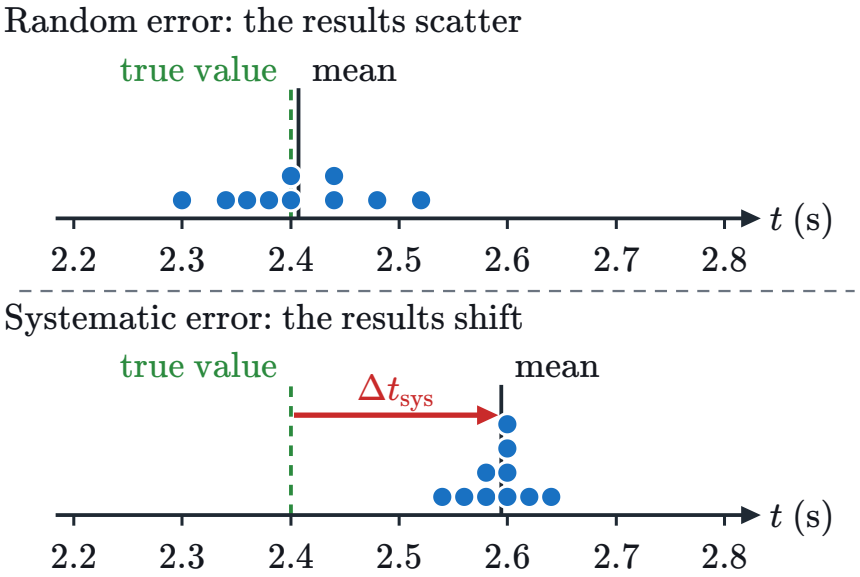
Random error

A varies from one measurement to the next by chance and can be in either direction. When the rolling time of a ball is measured with a hand stopwatch, it comes from reaction time: sometimes the stopwatch is started a little too early and sometimes too late. Reading a scale

between two marks also brings random error. It shows up as scatter: a slightly different number is obtained for the same quantity each time. Repeating the measurement makes the scatter visible, and in the mean the effect of random error is damped, although the scatter of the individual results does not get smaller: results that are too large and too small partly cancel each other.

Systematic error

A is almost the same in direction and size in every measurement. It comes, for example, from a wrong zero (a scale shows $1.2\ \mathrm{g}$ when empty), from a badly adjusted instrument, or from the way of measuring, such as looking at a scale from the side. All results shift in the same direction, so they can be in a tight group but in the wrong place. Repeating and the mean do not remove it. It is noticed by comparing the result with a known reference value or with another method of measurement, and then it is corrected.



Repeatability and accuracy

The reliability of a measurement is described by two things. tells how close to each other repeated measurements land, and it is weakened by random error. **Accuracy** tells how close the mean is to the true value, and it is weakened by systematic error. A measurement can be repeatable but not accurate, like the lower series in the figure. A reliable measurement has both.



The LIGO gravitational-wave detector at Hanford, USA. Its arms, kilometres long, detect changes in length far smaller than an atomic nucleus.

Photo: NOIRLab/LIGO/NSF/AURA/T. Matsopoulos, CC BY 4.0

Mean and error estimate

The best estimate of a repeated measurement is the mean of the results:

$$\bar{x} = \frac{x_1 + x_2 + \cdots + x_n}{n}$$

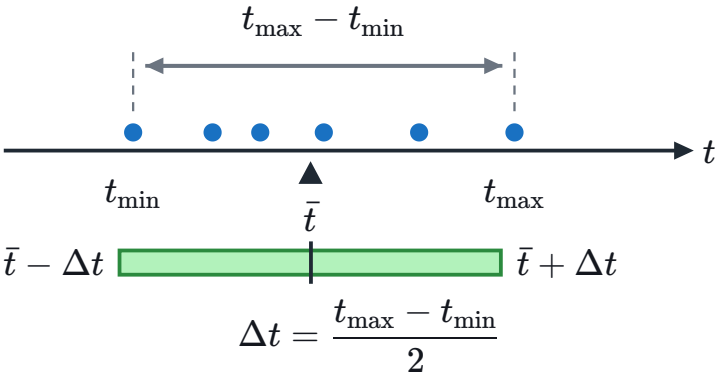
where $\bar{x}$ is the mean, $x_1, x_2, \ldots, x_n$ are the individual measurement results and n is the number of measurements. The mean adds up the

results and divides the sum by the number, so the more measurements there are, the less a single stray result affects it.

In this course the scatter of repeated results, that is, the size of random error, is estimated with a simple approximate method, as half of the range:

$$\Delta x = \frac{x_{\max} - x_{\min}}{2}$$

where Δx is the error estimate, $x_{\max}$ is the largest and $x_{\min}$ the smallest result. The error estimate, or **measurement uncertainty**, is not the measurement error itself, which is not known, but an estimate of how large the measurement error can be; it is always positive. The result is stated in the form $\bar{x} \pm \Delta x$. The error estimate is a rough estimate, not a strict limit: a single result can sometimes fall outside it. Half of the range describes the scatter of the individual results. The uncertainty of the mean is usually smaller, and the more measurements there are, the smaller it is, so for the mean Δx is a cautious estimate. The method suits the measurements of this course with a few repeats; more exact methods are based on statistical scatter.



Half of the range is a simple estimate of the scatter of the results, not a general method for finding the measurement uncertainty. It does not reveal a systematic error, which shifts all results in the same direction. If all the results are the same, no scatter shows, but the uncertainty is still not zero: then the reading uncertainty is estimated, as below.

The error estimate is rounded to one significant figure, or to two if the first figure is 1. The result is rounded to the same decimal place: $3.18\,\mathrm{s} \pm 0.12\,\mathrm{s}$ and not $3.1817\,\mathrm{s} \pm 0.12\,\mathrm{s}$. This way the result has no figures that cannot be trusted because of the uncertainty.

When the measurement is made only once and the value is read from a scale, in a simple situation the uncertainty of the reading can be estimated, for example, as half of the smallest scale division d:

$$\Delta x = \frac{d}{2}$$

With a ruler whose smallest scale division is $1.0\,\mathrm{mm}$, the uncertainty of the reading can thus be estimated as $0.5\,\mathrm{mm}$. However, the total uncertainty of a measurement can also be affected by the method of measurement, the instrument and other sources of error. The smallest scale division, or the step of the last figure of a display, is the **resolution** of the instrument, and it alone does not give the uncertainty: a stopwatch shows hundredths of a second, but the uncertainty of a time taken by hand is, because of reaction time, of the order of tenths of a second.

The ratio of the error estimate, that is, the measurement uncertainty, to the result is called the relative uncertainty:

$$\varepsilon = \frac{\Delta x}{\bar{x}} \cdot 100\,\%$$

where ε is the relative uncertainty as a percentage. It tells whether the uncertainty is small or large compared with the size of the result: an

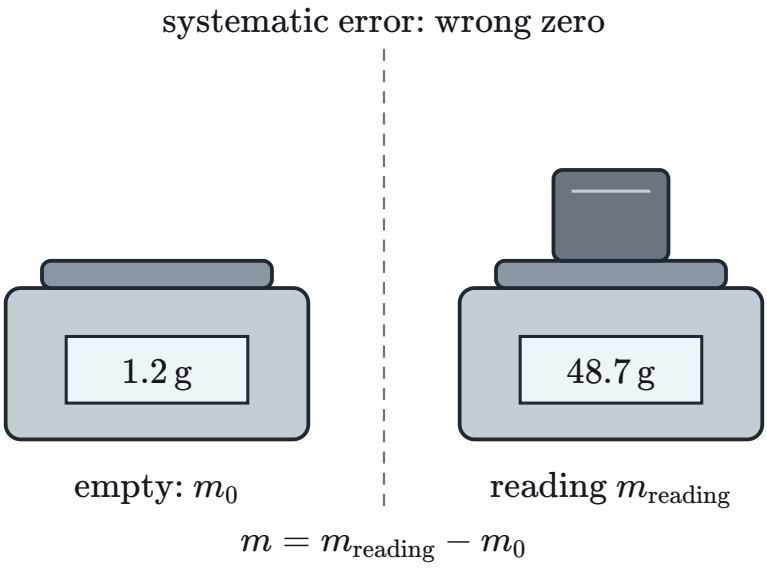
uncertainty of $0.5\ \mathrm{mm}$ is small for measuring a metre but large for a wire one millimetre thick.

A systematic error is estimated by measuring a quantity whose true value is known and comparing the mean of the measurements with this value. The reference can be, for example, an empty scale, ice water, whose temperature is $0\ ^\circ\mathrm{C}$, or a rod of precisely known length:

$$\Delta x_{\mathrm{sys}} = \bar{x}_{\mathrm{ref}} - x_{\mathrm{true}}$$

where $\bar{x}_{\mathrm{ref}}$ is the mean of the reference measurements and x_{true} is the true value of the reference quantity. The corrected result is obtained by subtracting the error from the reading:

$$x = x_{\mathrm{reading}} - \Delta x_{\mathrm{sys}}$$



Common misconception

Many people think that a precise instrument or many repeats guarantee a correct result. Even a precise instrument can be badly adjusted, and then its readings are in a tight group but in the wrong place. Repeats reduce random error but not systematic error, and good repeatability tells nothing about accuracy. Another misconception is to regard all error as carelessness. Carelessness causes a gross error, which is looked for and left out only for a justified and recorded reason, but random and systematic error belong to the result of a skilled measurer too, and that is why they are estimated and stated. A third is to calculate a mean of numbers that are in different units: the results are converted to the same unit first.

Summary

- A measurement error is the difference between the result and the true value, $x - x_{\text{true}}$. Every measurement result has some uncertainty. The error is usually not known, but its possible size is estimated as the measurement uncertainty and stated together with the result.
- A gross error is a single mistake, such as a wrongly read or recorded reading. It often shows up as a result that clearly differs from the others, which is checked; the result is left out only if the cause of the error is found, and the cause is recorded.
- A random error varies from one measurement to the next and shows up as scatter. In the mean $\bar{x}$ the effect of random error is damped, although the scatter of the individual results does not get smaller. In this course the scatter is estimated approximately by half of the range Δx, which is a cautious uncertainty estimate for the mean.
- A systematic error shifts all results in the same direction, and repeating does not remove it. It is noticed with a reference value and corrected: $x = x_{\text{reading}} - \Delta x_{\text{sys}}$.

- Repeatability tells about random error and accuracy about systematic error. A reliable measurement is both repeatable and accurate.
- The result is stated in the form $\bar{x} \pm \Delta x$ with its unit. The error estimate is rounded to one significant figure (two if the first figure is 1), and the result to the same decimal place.

Examples

Once you have studied the chapter's theory, move on to the guided examples. In the first you follow a worked solution step by step; in the next ones you make the choices yourself, and the hints help if you get stuck.

Example 1: Rolling time and its uncertainty

A student measures the rolling time of a ball on a ramp with a hand stopwatch six times and gets the results $3.12\ \text{s}$, $3.25\ \text{s}$, $3.07\ \text{s}$, $3.31\ \text{s}$, $3.19\ \text{s}$ and $3.15\ \text{s}$. What is the measurement result with its error estimate, and how large is the relative uncertainty?

STEP 1/7

Given and find. The number of measurements is $n = 6$, and the smallest result is $t_{\min} = 3.07\ \text{s}$ and the largest $t_{\max} = 3.31\ \text{s}$. The mean $\bar{t}$, the error estimate Δt and the relative uncertainty ε are asked for.

STEP 2/7

Principle. Hand timing has a random error, because reaction time varies. So the mean is taken as the result and half of the range as the error estimate. The relative uncertainty compares the error estimate with the result.

STEP 3/7

Rearranged formula.

$$\bar{t} = \frac{t_1 + t_2 + \cdots + t_n}{n}$$
$$\Delta t = \frac{t_{\max} - t_{\min}}{2}$$
$$\varepsilon = \frac{\Delta t}{\bar{t}} \cdot 100\,\%$$

STEP 4/7

Unit conversions. None needed: all times are in seconds, and the relative uncertainty is a number without a unit, which is multiplied by a hundred to get percent.

STEP 5/7

Substitution. The sum of the results is

$$3.12\,\mathrm{s} + 3.25\,\mathrm{s} + 3.07\,\mathrm{s} + 3.31\,\mathrm{s} + 3.19\,\mathrm{s} + 3.15\,\mathrm{s} = 19.09\,\mathrm{s}$$

Substitute it and the other numerical values with their units:

$$\begin{aligned}\bar{t} &= \frac{t_1 + t_2 + \cdots + t_n}{n} \\ &= \frac{19.09\,\mathrm{s}}{6} \\ &= 3.1817\,\mathrm{s}\end{aligned}$$

$$\begin{aligned}\Delta t &= \frac{t_{\max} - t_{\min}}{2} \\ &= \frac{3.31\,\mathrm{s} - 3.07\,\mathrm{s}}{2} \\ &= 0.12\,\mathrm{s}\end{aligned}$$

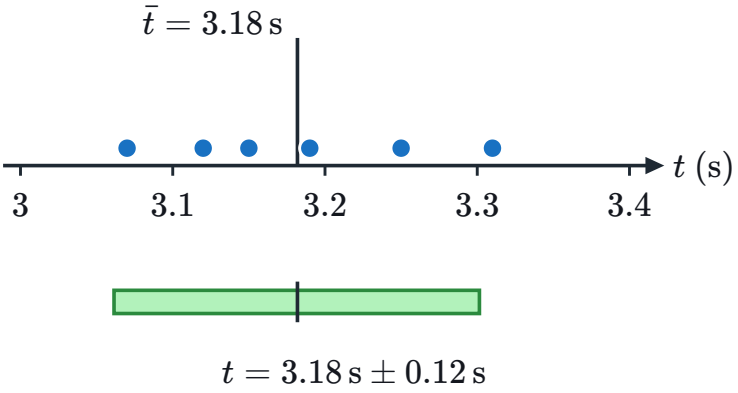
$$\begin{aligned}\varepsilon &= \frac{\Delta t}{\bar{t}} \cdot 100\,\% \\ &= \frac{0.12\,\mathrm{s}}{3.1817\,\mathrm{s}} \cdot 100\,\% \\ &= 3.8\,\%\end{aligned}$$

STEP 6/7

Result. The error estimate $0.12\,\mathrm{s}$ is a number with two figures, because its first figure is 1. The mean is rounded to the same decimal place, so the measurement result is $t = 3.18\,\mathrm{s} \pm 0.12\,\mathrm{s}$. The relative uncertainty is $3.8\,\%$, two significant figures like the error estimate.

STEP 7/7

Reasonableness. The interval $3.18\,\mathrm{s} \pm 0.12\,\mathrm{s}$, that is, $3.06\,\mathrm{s}$–$3.30\,\mathrm{s}$, covers five of the six results. The result $3.31\,\mathrm{s}$ falls just outside it, which fits the fact that the error estimate is rough. The relative uncertainty is a few percent, which is usual for a hand-timed measurement. However, the repeats tell only about random error: a systematic error would be revealed only by comparison with another method of measurement.



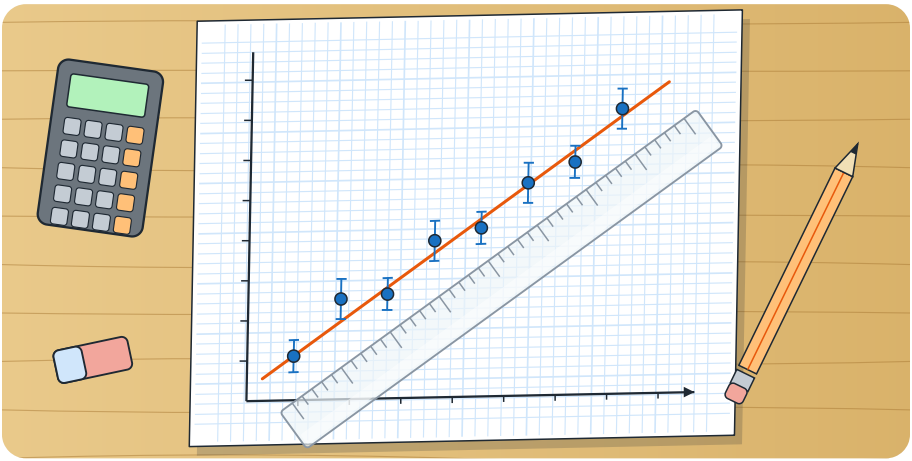
$$t = 3.18\,\mathrm{s} \pm 0.12\,\mathrm{s}$$

FY1 · CHAPTER 4

Graphing and fitting a straight line

Why do data points not usually all land exactly on one straight line, and yet a single line fitted to them can reveal the relationship between the quantities?

Simulation, tasks and model solutions online:
`ruukkilab.fi/en/fy1-4.html`



IN THIS LESSON, YOU WILL LEARN TO

- present measurements as a graph and fit a straight line to the points.
- find the slope and constant term of a line and say what they mean.
- recognise when a linear model does not fit the measurements.

PRIOR KNOWLEDGE

- Every measurement contains random error, which scatters the results in both directions
  chapter *Measuring and the reliability of a measurement*
- A quantity is given as a value and a unit, and the quantity and its unit are written on the axes of a graph
  chapter *Quantities, units and the SI system*

Core path: follow the steps in order. Besides calculations, the core path includes investigations, explanations and graphs. Tasks marked *Optional* give extra practice. Optional tasks: 4.

Think about it

The position of a trolley is measured on a track every second, and the results are plotted on a graph. The points lie roughly along a straight line, but no line passes through all of them. Two students draw different lines,

and each claims that their line shows the speed of the trolley.

> **How do you decide which line describes the measurements best, and what does the slope of the line tell you?**

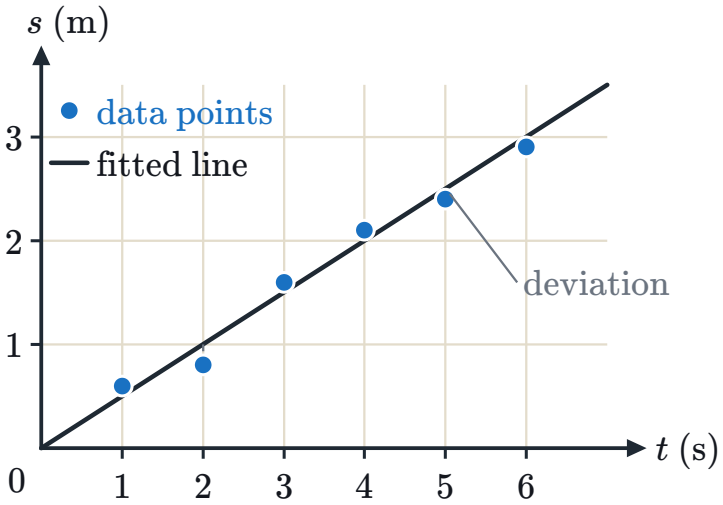
Theory

Key idea

A graph brings the measurements together

When you measure two quantities, for example the distance s travelled by a trolley at different times t, each measurement is one point on the graph. The horizontal coordinate of the point is the value of one quantity and the vertical coordinate is the value of the other. The horizontal axis usually gets the quantity that the experimenter chooses or that increases steadily, such as time, and the vertical axis gets the quantity being measured.

A graph needs axes with the quantity and its unit written at the end, such as $s\ (\mathrm{m})$, an evenly spaced scale and tick marks that are close enough together. Choose the scale so that the points fill most of the graph.



Fit a line, do not join the points

In real data sets the data points do not usually lie exactly on one straight line, because of measurement uncertainty, especially random error (chapter *Measuring and the reliability of a measurement*). So you do not join the points into a broken line, since that line would follow the errors too. Instead you fit one straight line that runs through the middle of the set of points.

A good fit is close to all the points and does not lean towards any single point. There are roughly as many points on each side of the line, and the deviations above and below roughly cancel each other. The line does not have to pass through any point. If one point is clearly further from the line than the others, it is worth finding out why, and you do not drop the point automatically.

Linear model

If the points lie roughly along a straight line, the relationship between the quantities is described by a linear model, whose equation is the equation of a straight line. The slope tells you how much the vertical quantity

changes when the horizontal quantity increases by one unit, and the constant term tells you the value of the vertical quantity when the horizontal quantity is zero. A linear model is a physical model, a simplified description of a phenomenon based on measurements: the graph is its graphical model and the equation of the line its mathematical model, and because the equation gives numerical values, the model is quantitative.

When the constant term is zero, the line passes through the origin and the quantities are directly proportional: doubling x doubles y. The measurements support the model only in the range that was measured, so estimating a value between the measured points (interpolation) is justified. If the line is extended beyond the measured range (extrapolation), you assume that the relationship stays the same, and the result is only an estimate. If the points curve systematically around the line, for example the points at the ends are above the line and the points in the middle below it, the linear model does not fit.

Equation of a straight line

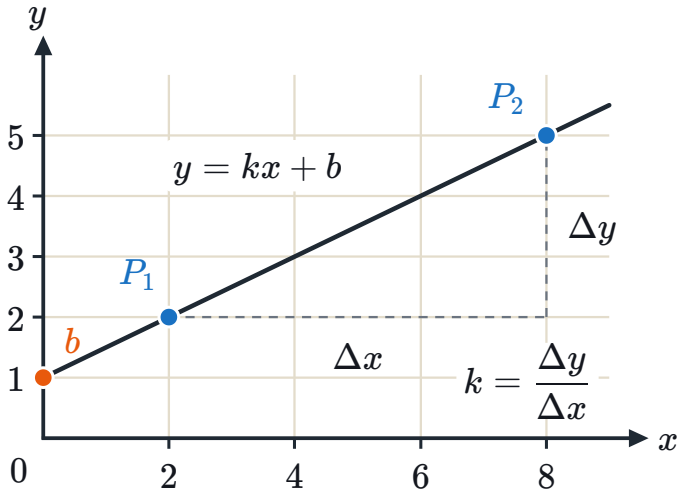
The graph of a linear model is a straight line with the equation

$$y = kx + b$$

where y is the quantity on the vertical axis, x the quantity on the horizontal axis, k the slope and b the constant term, that is, the point where the line crosses the vertical axis. The slope is calculated from two chosen points $(x_1,\ y_1)$ and $(x_2,\ y_2)$ on the line:

$$k = \frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1}$$

where Δy is the change in y between the points and Δx the change in x between them. The unit of the slope is the unit of y divided by the unit of x: if y is in metres and x in seconds, k is in $\mathrm{m/s}$.



Two things affect the result. Read the points from the fitted line and not from the data points, and choose them far apart, so that the reading error has only a small effect on the slope. Read the constant term b from the point where the line crosses the vertical axis, or calculate it from one point on the line: $b = y_1 - kx_1$. If you want x when y is known, solve the equation: $x = \frac{y-b}{k}$.

The sign of the slope tells you the direction. With a positive slope the line rises, and with a negative slope it falls. The slope usually has a physical meaning: in a distance–time graph it is the speed, and in a mass–volume graph it is the density.

Common misconception

Many students join the data points with a line, or force the line to pass through one of the data points, even though that point, too, has

measurement uncertainty. The line is fitted to the whole set of points; it may pass through some point, but it does not have to.

Another mistake is to calculate the slope from two data points. Then the error in each point affects the result, and if the points are also close together, the result is uncertain. Read the points from the fitted line and far apart.

A third mistake is to assume that the line always passes through the origin. If the constant term is not zero, the quantities are not directly proportional, and b tells you what y is when x is zero.

Summary

- The axes of a graph show the quantities with their units and an evenly spaced scale. The data points are not joined to each other.
- A straight line is fitted to the data points so that it runs through the middle of the set: there are roughly as many points on each side of it.
- The equation of the linear model is $y = kx + b$. The slope is $k = \frac{\Delta y}{\Delta x}$, and it is calculated from two points on the line that are far apart.
- The constant term b is the point where the line crosses the vertical axis. If $b = 0$, the quantities are directly proportional.
- If the points curve systematically, the linear model does not fit.

Examples

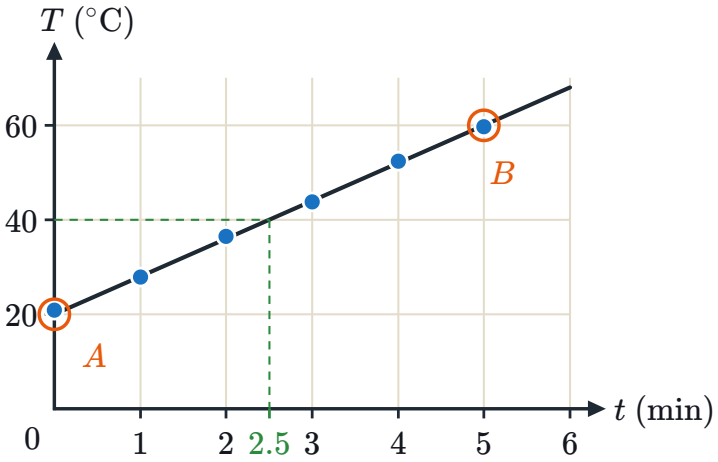
Once you have studied the chapter's theory, move on to the guided examples. In the first you follow a worked solution step by step; in the next ones you make the choices yourself, and the hints help if you get stuck.

Example 1: Fitting a line to kettle data

The temperature T of the water in a kettle was measured every minute:

t (min)	T (°C)
0	20.8
1	27.9
2	36.5
3	43.8
4	52.4
5	59.6

Fit a straight line to the data points, find its equation $T = kt + b$ and use it to work out when the water temperature is $40\,^{\circ}\mathrm{C}$. The power of the kettle is constant, and the heat losses are small.



STEP 1/7

Given and find. The graph has six data points, and the fitted line has been drawn through the middle of the set. Two points far apart are read from the line: $A = (t_1; T_1) = (0\,\text{min}; 20\,^\circ\text{C})$ and $B = (t_2; T_2) = (5.0\,\text{min}; 60\,^\circ\text{C})$. We need the slope k, the constant term b and the time t at which $T = 40\,^\circ\text{C}$.

STEP 2/7

Principle. The power of the kettle is constant and the heat losses are small in this temperature range, so the water temperature rises almost steadily and the linear model $T = kt + b$ fits. The line is drawn so that there are points on both sides of it and each point is close to the line. The slope is calculated from the points A and B on the line, not from the data points.

STEP 3/7

Rearranged formula. The slope comes from the differences between the points, the constant term from the point A, and the time is solved from the equation of the line:

$$k = \frac{T_2 - T_1}{t_2 - t_1}$$
$$b = T_1 - k t_1$$
$$t = \frac{T - b}{k}$$

STEP 4/7

Unit conversions. None needed: the temperatures are in degrees Celsius and the times in minutes. The unit of the slope is $^\circ\mathrm{C}/\mathrm{min}$, and a temperature change is the same in degrees Celsius and in kelvins.

STEP 5/7

Substitution.

$$
\begin{aligned}
k &= \frac{T_2 - T_1}{t_2 - t_1} \\
&= \frac{60\,^\circ\mathrm{C} - 20\,^\circ\mathrm{C}}{5.0\,\mathrm{min} - 0\,\mathrm{min}} \\
&= 8.0\,\tfrac{^\circ\mathrm{C}}{\mathrm{min}}
\end{aligned}
$$

$$
\begin{aligned}
b &= T_1 - kt_1 \\
&= 20\,^\circ\mathrm{C} - 8.0\,\tfrac{^\circ\mathrm{C}}{\mathrm{min}} \cdot 0\,\mathrm{min} \\
&= 20\,^\circ\mathrm{C}
\end{aligned}
$$

$$
\begin{aligned}
t &= \frac{T - b}{k} \\
&= \frac{40\,^\circ\mathrm{C} - 20\,^\circ\mathrm{C}}{8.0\,\tfrac{^\circ\mathrm{C}}{\mathrm{min}}} \\
&= 2.5\,\mathrm{min}
\end{aligned}
$$

STEP 6/7

Result. The equation of the line is

$$T = 8.0\,\tfrac{^\circ\mathrm{C}}{\mathrm{min}} \cdot t + 20\,^\circ\mathrm{C}$$

and the temperature $40\,^\circ\mathrm{C}$ is reached after about 2.5 minutes. The results are given to two significant figures, because k is determined from points read to two figures.

STEP 7/7

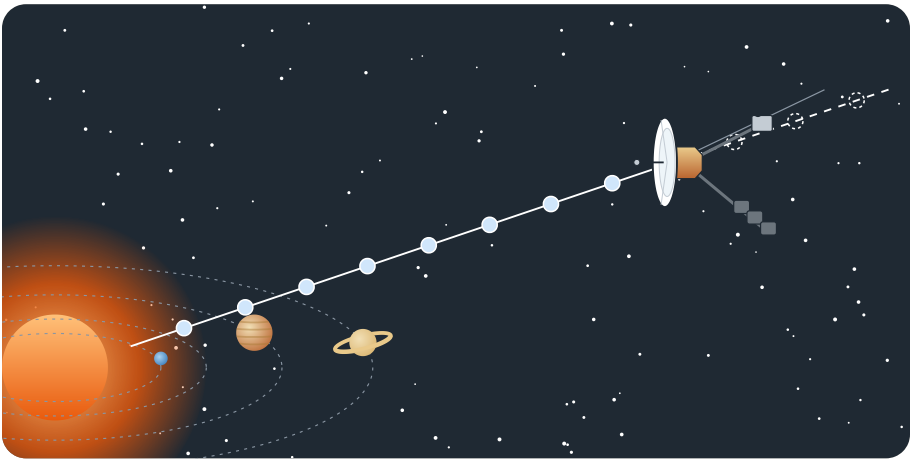
Reasonableness. The constant term $20\,^{\circ}\mathrm{C}$ is the temperature of the water at the start of heating according to the model, and it is about room temperature. The slope $8.0\,\frac{^{\circ}\mathrm{C}}{\mathrm{min}}$ tells you how fast the water warms up. In the table, $40\,^{\circ}\mathrm{C}$ is passed between the third and fourth measurements, since T is $36.5\,^{\circ}\mathrm{C}$ at two minutes and $43.8\,^{\circ}\mathrm{C}$ at three minutes, so $2.5\,\mathrm{min}$ is in the right interval. The points deviate from the line by at most about one degree.

FY1 · CHAPTER 5

Uniform motion: speed, position and graphs

Why can the position of the Voyager 2 probe be calculated in advance for any day, even though it is more than 20 billion kilometres from the Sun?

Simulation, tasks and model solutions online:
`ruukkilab.fi/en/fy1-5.html`



IN THIS LESSON, YOU WILL LEARN TO

- calculate the distance, time or speed in uniform motion with the formula $s = s_0 + vt$.
- read the speed and the initial position from a (t, s) graph.
- explain when the distance is directly proportional to time and when the position is.

PRIOR KNOWLEDGE

- A straight line is fitted to measurement points, and its slope tells you how much y changes when x increases by one
  chapter *Graphing and fitting a straight line*
- In the equation of a straight line, the constant term is where the line crosses the y-axis
  chapter *Graphing and fitting a straight line*

Core path: follow the steps in order. Besides calculations, the core path includes investigations, explanations and graphs. Tasks marked *Optional* give extra practice. Optional tasks: 3.

Think about it

The Voyager 2 probe left the Earth in 1977. In November 2018 it crossed the heliopause, the boundary beyond which the solar wind no longer dominates space.

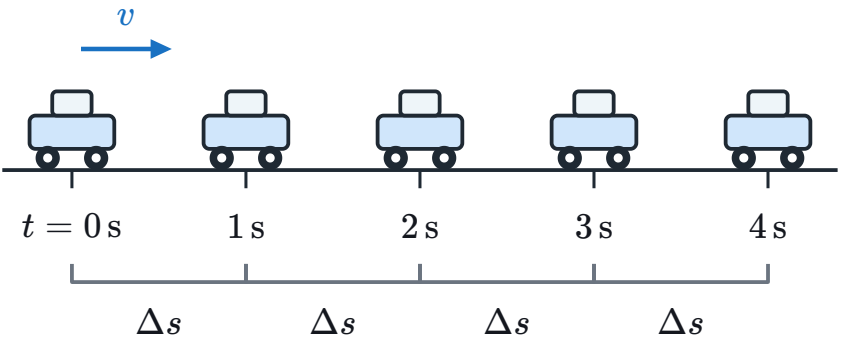
According to the measurements, the probe was then $119.0\ \mathrm{AU}$ from the Sun (Burlaga et al. 2019), and it was moving away at a speed of $15.341\ \mathrm{km/s}$ (NASA/JPL). The astronomical unit AU is roughly the distance between the Earth and the Sun. You cannot hold a ruler next to the probe, but you can still give its distance for any day.

> **What information do you need to calculate the distance of the probe from the Sun, say a year from now, and what do you assume about its speed?**

Theory

Key idea

Uniform motion is motion along a straight line at constant speed: in equal intervals of time, the object travels equal distances. Speed tells you how far the object travels in one second, so its unit is metres per second (m/s). If a trolley moves two metres every second, its speed is $2\ \mathrm{m/s}$.



equal distances in equal intervals of time



With a long exposure the lights of cars draw lines in the picture: the white ones are headlights and the red ones rear lights. The photo records the path of the motion, but it does not show whether the motion was uniform.

Photo: Tao Yuan, Unsplash License

Direct proportionality

Two quantities are directly proportional if their ratio is always the same. When uniform motion starts from the zero point, the position s, which

here equals the distance travelled, and the time t are directly proportional: double the time gives double the distance, and the ratio $s : t$ is the speed v. In a (t, s) graph this shows as a straight line through the origin.

Linear model

Uniform motion is a model, not reality. A cyclist pedals unevenly, and Voyager 2 slows down slightly because of the Sun's gravity. The model is still good when the speed changes only a little during the time being considered. Its graph is a straight line, and a straight line is described by a linear model: the position changes by the same amount every second. Values calculated with the model are model values, which are compared with measurements.

Uniform motion

Suppose that at the start of the observation ($t = 0$) an object is at position s_0 and moves along a straight line at constant speed v. Its position at time t is

$$s = s_0 + vt$$

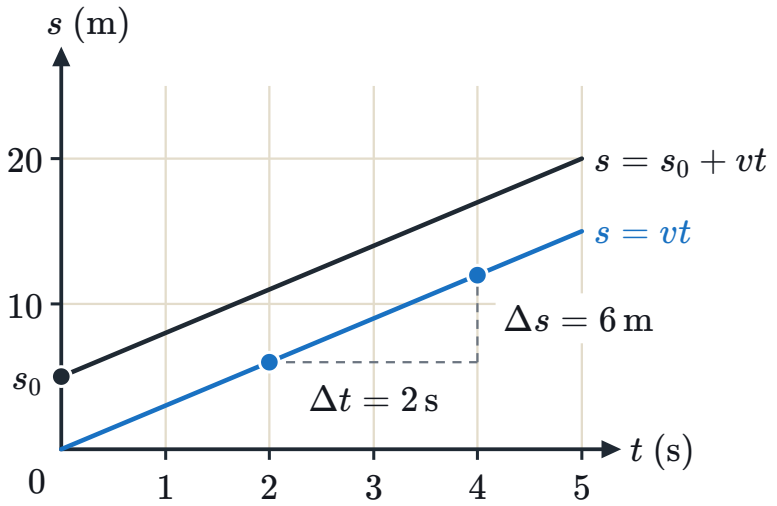
where s is the position (m) measured from the chosen zero point, s_0 the initial position (m), v the speed (m/s) and t the time from the start (s). The distance travelled by the object is $\Delta s = s - s_0 = vt$. In the examples of this chapter the object moves in the positive direction, so the velocity is positive.

Speed from a graph

The (t, s) graph is a straight line, and the slope of the line is the speed:

$$v = \frac{\Delta s}{\Delta t}$$

where Δs is the change in position (m) and Δt is its time interval (s). The line crosses the s-axis at the initial position s_0. If $s_0 = 0$, the position is directly proportional to time: $s = vt$. If $s_0 \neq 0$, the line is as steep as the line through the origin at the same speed, but shifted up or down. Then only the distance Δs, not the position s, is directly proportional to time.



Units

Always convert to compatible units (usually SI units) before substituting: substitute the value of the unit in its place. For example $1\,\mathrm{km} = 1000\,\mathrm{m}$ and $1\,\mathrm{h} = 3600\,\mathrm{s}$, so $72\,\mathrm{km/h} = \frac{72 \cdot 1000\,\mathrm{m}}{3600\,\mathrm{s}} = 20\,\mathrm{m/s}$. In the same way $1\,\mathrm{min} = 60\,\mathrm{s}$. Distances on the scale of the Solar System are often given in astronomical units: $1\,\mathrm{AU} = 149\,597\,870.7\,\mathrm{km}$ is roughly the mean distance between the Earth and the Sun.

Common misconception

Many people think that in uniform motion the position doubles when the time doubles. This happens only if the motion starts from the zero point. When a trolley starts at $s_0 = 6\,\mathrm{m}$ and moves at $2.0\,\mathrm{m/s}$, it is at $10\,\mathrm{m}$ at $2.0\,\mathrm{s}$ and at $14\,\mathrm{m}$ at $4.0\,\mathrm{s}$, not at $20\,\mathrm{m}$. The distance, however, does double: $4.0\,\mathrm{m}$ becomes $8.0\,\mathrm{m}$. Another mistake is to calculate the speed as the quotient of position and time $s : t$ instead of the quotient of the change in position and the time interval: the speed is $\Delta s : \Delta t$, not $s : t$, when $s_0 \neq 0$.

Summary

- In uniform motion the speed is constant, and the object travels equal distances in equal intervals of time: $s = s_0 + vt$.
- The distance travelled $\Delta s = vt$ is directly proportional to time. The position s is proportional to time only when $s_0 = 0$.
- The (t, s) graph is a straight line. Its slope $\Delta s : \Delta t$ is the speed, and the intercept on the s-axis is the initial position s_0.
- Convert units before calculating by substitution, for example $1\,\mathrm{km} = 1000\,\mathrm{m}$ and $1\,\mathrm{h} = 3600\,\mathrm{s}$.
- Uniform motion is a linear model. The values it gives are model values, which can be compared with measurements, such as the distance of Voyager 2.

Examples

Once you have studied the chapter's theory, move on to the guided examples. In the first you follow a worked solution step by step; in the next ones you make the choices yourself, and the hints help if you get stuck.

Example 1: Voyager 2's distance

Voyager 2 crossed the heliopause on 5 November 2018 at a distance of $s_0 = 119.0\ \mathrm{AU}$ from the Sun (Burlaga et al. 2019, Nature Astronomy). The speed of the probe relative to the Sun at the crossing was $v = 15.341\ \mathrm{km/s}$ (NASA/JPL). Assume the speed is constant. According to the model, how far from the Sun is the probe on 5 November 2026?

STEP 1/7

Given and find. $s_0 = 119.0\ \mathrm{AU}$ at time $t = 0$ (5 November 2018), $v = 15.341\ \mathrm{km/s}$ and $1\ \mathrm{AU} = 149\,597\,870.7\ \mathrm{km}$. The time interval from 5 November 2018 to 5 November 2026 is 8 years, including the leap days of 2020 and 2024: $8 \cdot 365 + 2 = 2922$ days. The quantity to find is the distance s from the Sun.

STEP 2/7

Principle. When the speed is assumed constant, the probe moves uniformly and its position follows the linear model. The result is a model value, not a measurement: the Sun's gravity slows the probe slightly, by only about $0.1\ \mathrm{km/s}$ over 2018–2026, which is negligible here.

STEP 3/7

Rearranged formula.

$$s = s_0 + vt$$

STEP 4/7

Unit conversions. The time is already in days, $t = 2922\,\text{days}$, so the speed is converted to AU/day. In place of the units we substitute $1\,\text{km} = \frac{1}{149\,597\,870.7}\,\text{AU}$ and $1\,\text{s} = \frac{1}{86\,400}\,\text{day}$:

$$\begin{aligned} v &= 15.341\,\tfrac{\text{km}}{\text{s}} \\ &= 15.341 \cdot \tfrac{1}{149\,597\,870.7}\,\text{AU} : \left(\tfrac{1}{86\,400}\,\text{day}\right) \\ &= 0.0088602\,\tfrac{\text{AU}}{\text{day}} \end{aligned}$$

STEP 5/7

Substitution.

$$\begin{aligned} s &= s_0 + vt \\ &= 119.0\,\text{AU} \\ &\quad + 0.0088602\,\tfrac{\text{AU}}{\text{day}} \cdot 2922\,\text{days} \\ &= 119.0\,\text{AU} + 25.89\,\text{AU} \\ &= 144.89\,\text{AU} \end{aligned}$$

STEP 6/7

Result. According to the model, the probe is $s \approx 144.9\,\text{AU}$ from the Sun, about $2.17 \cdot 10^{10}\,\text{km}$. The initial distance is given to the nearest tenth, so the result is rounded to the nearest tenth.

STEP 7/7

Reasonableness. A speed of $15\,\text{km/s}$ covers in a year about

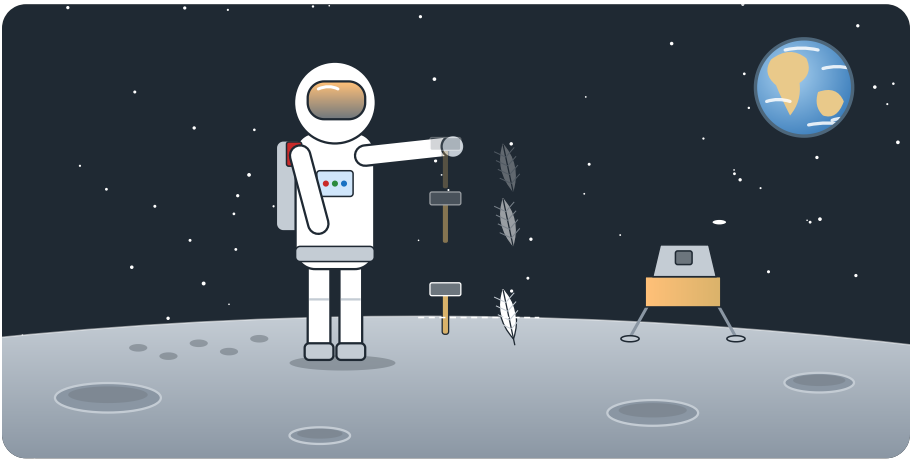
$$
\begin{aligned}
&15\,\mathrm{km/s} \cdot 3.16 \cdot 10^{7}\,\mathrm{s} \\
&\quad \approx 4.7 \cdot 10^{8}\,\mathrm{km} \\
&\quad \approx 3.2\,\mathrm{AU}
\end{aligned}
$$

so in 8 years the probe advances about $26\ \mathrm{AU}$ and $119.0\ \mathrm{AU} + 26\ \mathrm{AU} \approx 145\ \mathrm{AU}$. For 15 February 2026 the same model gives a distance from the Sun of $142.6\ \mathrm{AU}$. The NASA/JPL Voyager mission status page gave the probe's distance from the Earth in February 2026 as $143.05\ \mathrm{AU}$. The Earth orbits the Sun, so the probe's distance from the Earth differs from its distance from the Sun by up to about $1\ \mathrm{AU}$ over the year. The difference of $0.5\ \mathrm{AU}$ is smaller than this: the model and the observation agree to within about $1\ \mathrm{AU}$, and this comparison cannot test the model more precisely. That is less than one percent of the distance, so the constant-speed model suits predictions over a few years.

Acceleration of free fall: $v = g\,t$

Why did a hammer and a feather hit the Moon's surface together when the astronaut dropped them at the same time, even though on Earth a feather drifts slowly down?

Simulation, tasks and model solutions online:
`ruukkilab.fi/en/fy1-6.html`



IN THIS LESSON, YOU WILL LEARN TO

- calculate the speed of a freely falling object with the formula $v = g\,t$.
- read the acceleration of free fall from the slope of the (t, v) graph.
- conclude from experiments when mass affects falling and when it does not.

PRIOR KNOWLEDGE

- In uniform motion the distance is the product of speed and time, $s = vt$
  chapter *Uniform motion: speed, position and graphs*
- The slope of a straight-line graph tells how much the quantity on the vertical axis changes when the quantity on the horizontal axis increases by one
  chapter *Graphing and fitting a straight line*

Core path: follow the steps in order. Besides calculations, the core path includes investigations, explanations and graphs. Tasks marked *Optional* give extra practice. Optional tasks: 3.

Think about it

In 1971, on the Apollo 15 mission, astronaut David Scott held a hammer in one hand and a falcon feather in the

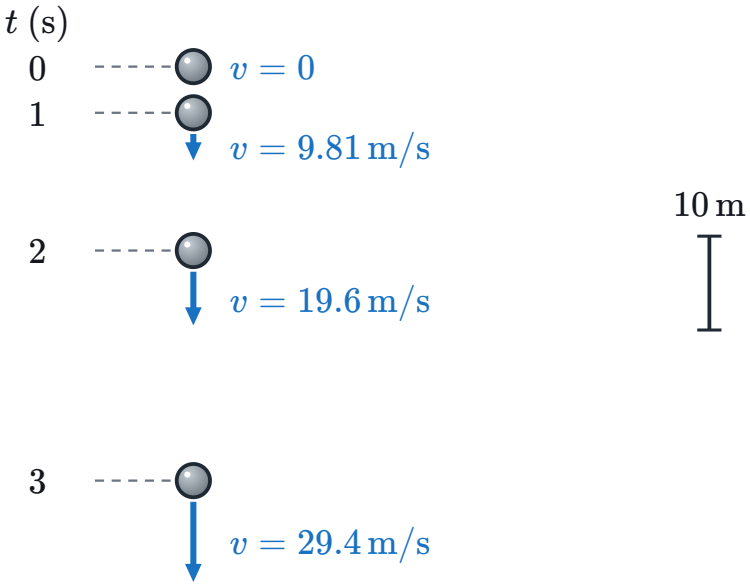
other on the surface of the Moon and let them go at the same time. They hit the lunar surface at the same moment. At home the same experiment does not work: the hammer falls straight down, but the feather drifts slowly.

> **Does the speed of a falling object increase steadily, and does a heavy object fall faster than a light one?**

Theory

Key idea

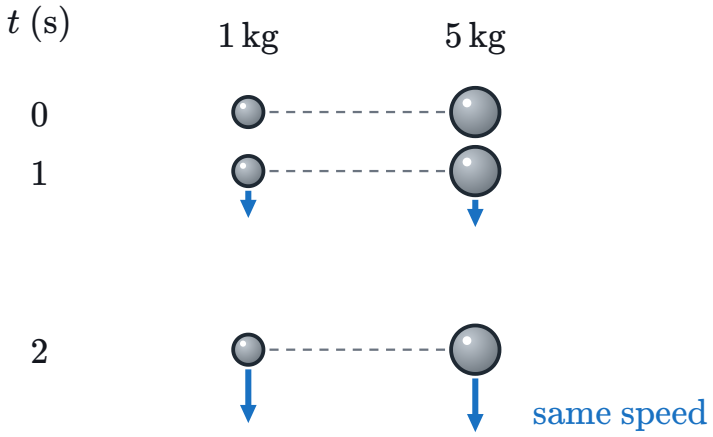
When an object is released from rest, its speed increases. is falling with no air resistance. Measurements show that in free fall the speed increases by the same amount every second: on the surface of the Earth by $9.81\ \mathrm{m/s}$ per second. After one second the speed is $9.81\ \mathrm{m/s}$, after two seconds $19.6\ \mathrm{m/s}$ and after three seconds $29.4\ \mathrm{m/s}$. So the object does not fall at a constant speed as in uniform motion (chapter *Uniform motion: speed, position and graphs*) but accelerates.



The change in speed per unit of time is called . The acceleration of a freely falling object is the $g = 9.81\ \mathrm{m/s^2}$. The unit $\mathrm{m/s^2}$ is read "metres per second squared", that is, "metres per second per second": speed is measured in $\mathrm{m/s}$, and it changes by this amount every second. So the number g is not a speed but the rate at which the speed increases.

Mass does not affect falling

When there is no air resistance, all objects fall with the same acceleration regardless of their mass. In a vacuum a hammer and a feather fall equally fast, and a ball with a mass of $5\ \mathrm{kg}$ has at every moment the same speed as a ball with a mass of $1\ \mathrm{kg}$. The reason is explained later in connection with forces. In this chapter it is taken as an experimental result that you investigate yourself.



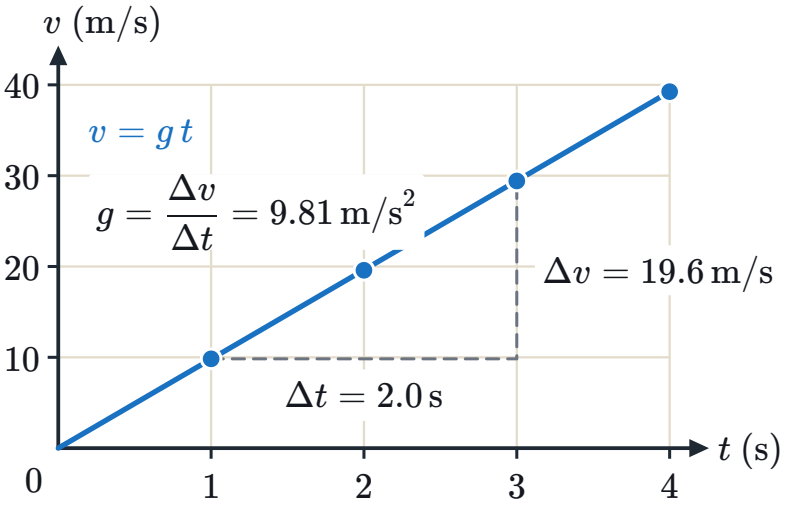
In air the situation is different. Air slows a light, wide object, such as a feather or a sheet of paper, clearly more than a small, dense object, such as a stone or a steel ball. This is why at home a feather falls more slowly than a hammer. The difference is caused by the air, not by mass as such.

Speed in free fall

Because the speed increases by the same amount every second, the speed is directly proportional to time. When the object starts from rest, its speed at the moment t is

$$v = g\,t$$

where v is the downward speed $(\mathrm{m/s})$, $g = 9.81\ \mathrm{m/s^2}$ is the acceleration of free fall and t is the time from the start of the fall (s). The formula is a linear model $v = kt$ in which the slope k is g. It holds when there is no air resistance.



So the (t, v) graph of a falling object is a rising straight line through the origin. Its slope $\frac{\Delta v}{\Delta t}$ is the acceleration of free fall, so g can be found from the graph by reading two points far apart. Because the acceleration is constant, $\Delta v = g\,\Delta t$ holds for any time interval: the speed of an object moving downwards increases by $9.81\ \mathrm{m/s}$ every second even when the object did not start from rest.



A bouncing ball photographed at equal time intervals. As the ball falls it speeds up, so the gaps between successive images grow.

Photo: MichaelMaggs, Richard Bartz, CC BY-SA 3.0

Common misconception

A heavy object does not fall faster than a light one: when there is no air resistance, all objects have the same g. Everyday experience seems to show otherwise, because air holds back the fall of a light, wide object the most. Another mistake is to treat the number $g = 9.81\ \mathrm{m/s^2}$ as a speed: it tells how much the speed increases every second, and the speed itself is $g\,t$.

Summary

- In free fall there is no air resistance. The speed of the object increases by the same amount every second.
- The acceleration of free fall on the surface of the Earth is $g = 9.81\ \mathrm{m/s^2}$, and it is the same for all objects regardless of mass.

- The speed of an object that starts from rest is $v = g\,t$, and the (t, v) graph is a straight line through the origin with slope g.
- In air, air holds back the fall of a light, wide object the most, so the formula $v = g\,t$ holds only when air resistance can be ignored.
- Conversion: at a speed of $1\ \mathrm{m/s}$ an object travels $3600\ \mathrm{m} = 3.6\ \mathrm{km}$ in an hour, so $1\ \mathrm{m/s} = 3.6\ \mathrm{km/h}$.

Examples

Once you have studied the chapter's theory, move on to the guided examples. In the first you follow a worked solution step by step; in the next ones you make the choices yourself, and the hints help if you get stuck.

Example 1: A stone dropped from a bridge

A stone is dropped from rest from a bridge, and it hits the water $1.5\ \mathrm{s}$ later. What is the speed of the stone when it hits the water, in $\mathrm{m/s}$ and in $\mathrm{km/h}$? Air resistance is ignored.

STEP 1/7

Given and find. The duration of the fall is $t = 1.5\ \mathrm{s}$, the stone starts from rest and the acceleration of free fall is $g = 9.81\ \mathrm{m/s^2}$. We want the speed v at the moment of impact, first in $\mathrm{m/s}$ and then in $\mathrm{km/h}$.

STEP 2/7

Principle. The stone is in free fall, because air resistance is ignored, and it starts from rest. Then the speed increases by the same amount every second, so the speed is directly proportional to time: $v = g\,t$.

STEP 3/7

Rearranged formula. The formula is already solved for the quantity we want:

$$v = g\,t$$

STEP 4/7

Unit conversions. The time is in seconds and g in $\mathrm{m/s^2}$, so the result comes in $\mathrm{m/s}$ without a conversion. To convert to kilometres per hour, think about what $1\ \mathrm{m/s}$ means: in an hour ($3600\ \mathrm{s}$) an object at this speed covers $3600\ \mathrm{m} = 3.6\ \mathrm{km}$, so $1\ \mathrm{m/s} = 3.6\ \mathrm{km/h}$. This is substituted in place of the unit.

STEP 5/7

Substitution.

$$\begin{aligned} v &= g\,t \\ &= 9.81\ \mathrm{m/s^2} \cdot 1.5\ \mathrm{s} \\ &= 14.715\ \mathrm{m/s} \end{aligned}$$

$$\begin{aligned} v &= 14.715 \cdot 1\ \mathrm{m/s} \\ &= 14.715 \cdot 3.6\ \mathrm{km/h} \\ &= 52.97\ \mathrm{km/h} \end{aligned}$$

STEP 6/7

Result. The speed is $v \approx 15\ \mathrm{m/s}$, that is, $v \approx 53\ \mathrm{km/h}$. Two significant figures, because the time $1.5\ \mathrm{s}$ is given with two figures. The conversion is made from the unrounded value, not from the rounded $15\ \mathrm{m/s}$.

STEP 7/7

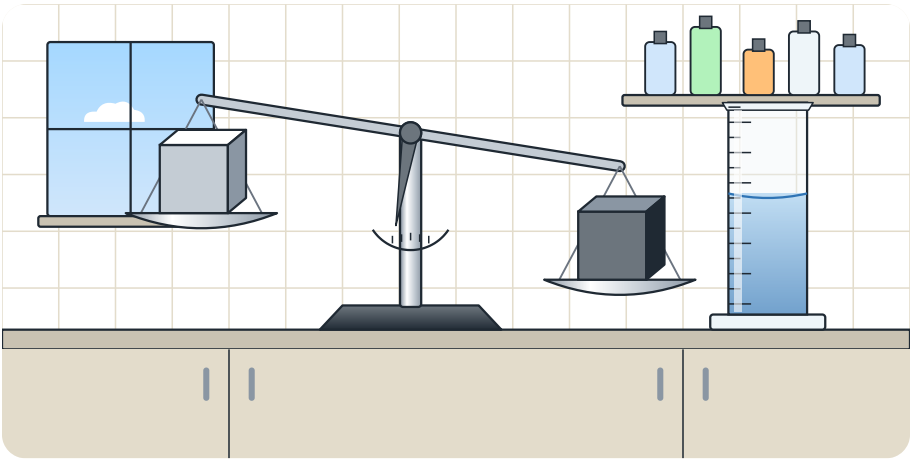
Reasonableness. The graph shows the same: at the moment $1\ \mathrm{s}$ the speed is $9.81\ \mathrm{m/s}$ and at the moment $2\ \mathrm{s}$ it is $19.6\ \mathrm{m/s}$, so at the moment $1.5\ \mathrm{s}$ it is about $15\ \mathrm{m/s}$. A speed of $53\ \mathrm{km/h}$ is a normal speed for a car in city traffic, and it is a reasonable result for a fall that lasts only one and a half seconds.

FY1 · CHAPTER 7

Density: $m = \rho V$

Why can two metal cubes of the same size feel completely different in weight in your hand, and how can you work out what an unknown object is made of using only a balance and a measuring cylinder?

Simulation, tasks and model solutions online:
`ruukkilab.fi/en/fy1-7.html`



IN THIS LESSON, YOU WILL LEARN TO

- calculate the mass, volume or density of an object with the formula $m = \rho V$.
- explain why mass is directly proportional to volume and what the slope of the line tells you.
- determine the density of a substance from measurement results and identify the substance using a table.

PRIOR KNOWLEDGE

- In direct proportionality the measurement points lie, within the measurement uncertainty, on a straight line through the origin, and its slope is the ratio $y : x$
  chapter *Graphs and fitting a straight line*
- Measurement results always contain error, and a result is given with a suitable number of significant figures
  chapter *Measuring and the reliability of a measurement*
- Quantities are calculated in SI units, and units are converted using powers of ten
  chapter *Quantities, units and the SI system*

Core path: follow the steps in order. Besides calculations, the core path includes investigations, explanations and graphs. Tasks marked *Optional* give extra practice. Optional tasks: 3.

Think about it

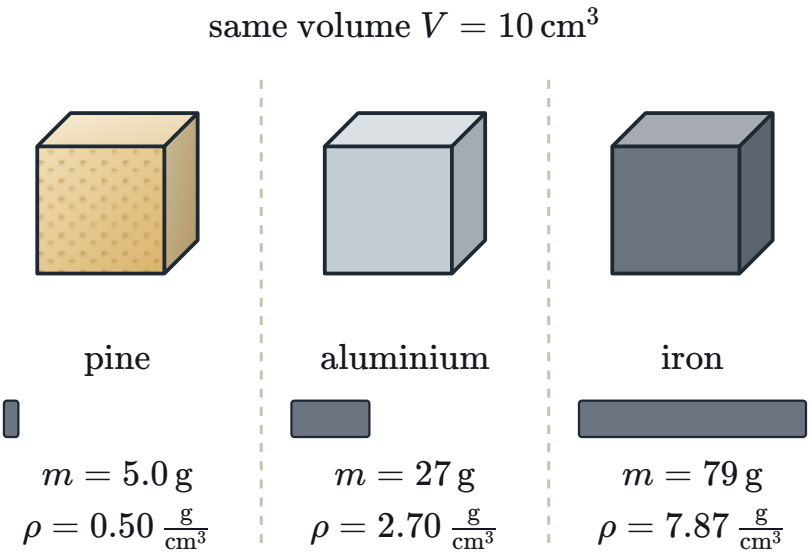
At a recycling point there are two grey metal cubes on the table. They are exactly the same size, but one feels much heavier in your hand than the other. The cubes have no markings, and nobody knows which metal they are made of. On the table there are a balance and a ruler.

> **Which quantities of the cubes should you measure so that you can work out what they are made of?**

Theory

Key idea

The mass m of an object tells you how much matter the object contains, and its unit is the kilogram. The volume V tells you how much space the object takes up; its SI unit is the cubic metre (m^3). Two objects of the same size may have different weights if they are made of different substances: a ten-cubic-centimetre pine cube is light, an aluminium cube is clearly heavier and an iron cube is the heaviest.



The difference comes from how tightly the matter is packed inside the volume. This is described by ρ: it tells you how much mass a substance has per unit of volume. Iron is about sixteen times as dense as pine, so the same volume contains about sixteen times as much matter.

Mass is directly proportional to volume

When you measure pieces of different sizes made of the same substance, you notice a regularity: twice the volume gives twice the mass, and half the volume gives half the mass. So mass is **directly proportional** to volume, which means the ratio $m : V$ stays the same. That constant ratio is the density of the substance.

If the measurement results are plotted in a coordinate system with volume on the horizontal axis and mass on the vertical axis, the points lie, within the measurement uncertainty, on a straight line through the origin (chapter *Graphs and fitting a straight line*). The slope of the line is the density.

Density tells you what the substance is

Every pure substance has its own density in a given state of matter and at a given temperature and pressure, so density is like a fingerprint of the substance: once the mass and volume of an object have been measured, the calculated density can be compared with table values. The conclusion is not certain. Measurement results always contain uncertainty (chapter *Measuring and the reliability of a measurement*), and some substances have almost the same density, so the result only tells you which substances the object may be made of.



An iceberg in a glacier lagoon in Iceland. Ice floats in water because its density is smaller than that of water.

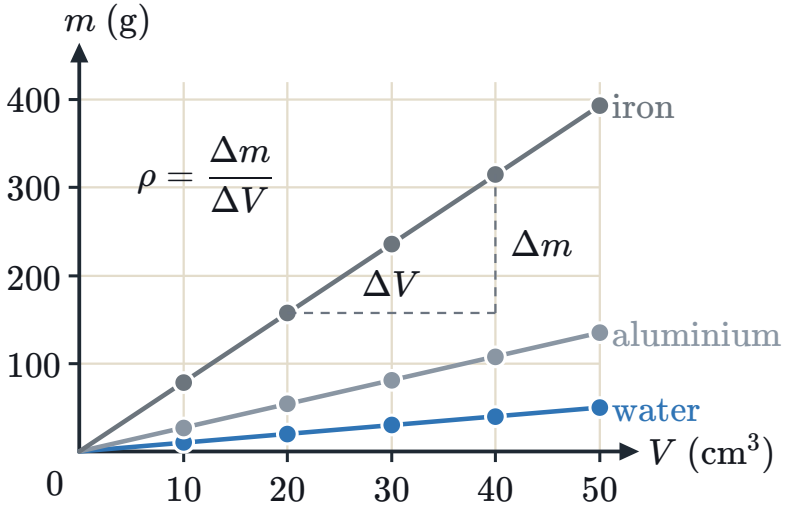
Photo: Eric Kilby, CC BY-SA 2.0

Density

The density of a substance is the ratio of mass to volume. Solving this equation for mass and for volume gives two other forms:

$$\rho = \frac{m}{V}$$
$$m = \rho V$$
$$V = \frac{m}{\rho}$$

where ρ is the density (SI unit $\mathrm{kg/m^3}$), m the mass (kg) and V the volume ($\mathrm{m^3}$). The form $m = \rho V$ is a model of direct proportionality $y = kx$: the mass m is y, the volume V is x and the density ρ is the slope k.



In the figure, the points of each substance lie on their own line. The steeper the line, the greater the density. You can get the density from a graph by reading two points far apart on the line and calculating $\rho = \Delta m : \Delta V$. The result obtained this way is more reliable than the ratio $m : V$ of a single measurement point, because the line has been fitted to all the measurement points.

Units

The SI unit of density is $\mathrm{kg/m^3}$, but for small objects the unit $\mathrm{g/cm^3}$ is often used. The relationship between the units is

$$1\ \tfrac{\mathrm{g}}{\mathrm{cm^3}} = \frac{10^{-3}\,\mathrm{kg}}{10^{-6}\,\mathrm{m^3}} = 1000\ \tfrac{\mathrm{kg}}{\mathrm{m^3}}$$

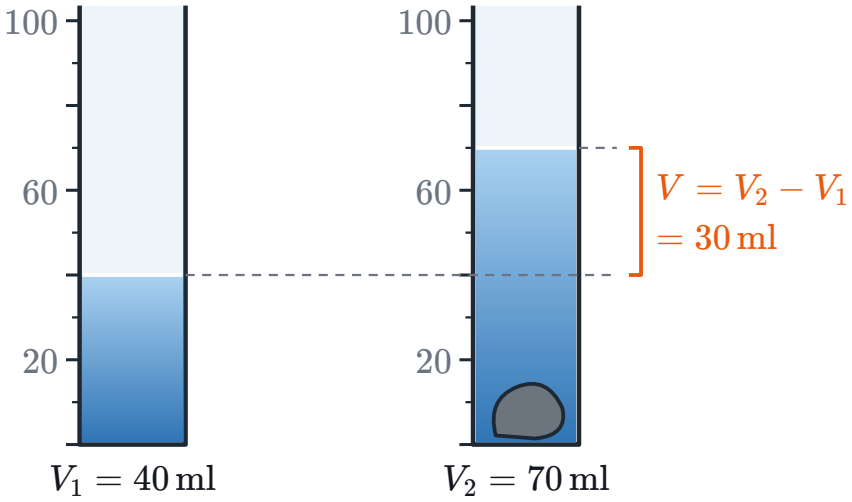
In a calculation, keep all quantities in the same system of units. The volume conversions are $1\,\mathrm{cm^3} = 10^{-6}\,\mathrm{m^3}$ and $1\,\mathrm{dm^3} = 10^{-3}\,\mathrm{m^3}$. In addition, $1\,\mathrm{dm^3}$ is one litre, and $1\,\mathrm{ml} = 1\,\mathrm{cm^3}$.

Substance	$\rho\,(\mathrm{kg/m^3})$	$\rho\,(\mathrm{g/cm^3})$
pine	500	0.50
ethanol	790	0.79
cooking oil	920	0.92
ice	920	0.92
water	1000	1.00
aluminium	2700	2.70
iron	7870	7.87
copper	8960	8.96
lead	11 300	11.3
gold	19 300	19.3

The values in the table are at room temperature, the value for ice at $0\,^\circ\mathrm{C}$. The density of wood varies with its moisture and quality, so the value for pine is a typical value.

Measuring volume

The volume of a regular object is calculated from its dimensions: the volume of a cube is a^3 and the volume of a rectangular block is the area of the base times the height. The volume of an irregular object is measured by displacement of water: the object is fully submerged in the water in a measuring cylinder, and its volume is the difference of the cylinder readings, $V = V_2 - V_1$.



A reading of a measuring cylinder contains uncertainty, for example because the scale is read at a curved liquid surface. The difference of two readings is a smaller number than the readings themselves, but it includes the uncertainty of both readings, so the difference may have fewer significant figures than either reading. This limits the accuracy of the density.

The mass of a container in a measurement

If a liquid is measured in a container and the balance shows the total mass of the container and the liquid, the line fitted to the measurement

points does not pass through the origin: the mass m_0 of the empty container is included. The total mass follows a linear model $m_{\text{tot}} = \rho V + m_0$. The density is still the slope of the line, and the constant term m_0 is the intercept of the line on the vertical axis, that is, the mass of the empty container. The total mass is not directly proportional to volume, but the mass of the liquid itself is.

Common misconception

Many people think that a bigger object is denser because it is heavier. A bigger piece of iron is indeed heavier than a small one, but their density is the same: mass and volume grow together so that the ratio $m : V$ does not change. Density is a property of the substance, not of the size of the object.

Another mistake is to mix up units. If the mass is in grams but the volume in cubic metres and the result is read in $\mathrm{kg/m^3}$, the number is a thousand times too large. First convert everything to one system of units: either kg and $\mathrm{m^3}$ or g and $\mathrm{cm^3}$.

A third mistake is to forget the mass of the empty container. From a single weighing it is subtracted before the density is calculated; from a graph the density is read from the slope.

Summary

- Density is mass per unit of volume: $\rho = m : V$, SI unit $\mathrm{kg/m^3}$. Water $1000\ \mathrm{kg/m^3}$, aluminium $2700\ \mathrm{kg/m^3}$, iron $7870\ \mathrm{kg/m^3}$.
- The mass of a given substance is directly proportional to volume: $m = \rho V$. In a graph, $m(V)$ is a straight line through the origin whose slope is the density.
- Solve the formula for the quantity you need: $\rho = m : V$, $m = \rho V$ or $V = m : \rho$. Keep the units in one system: $1\ \mathrm{g/cm^3} = 1000\ \mathrm{kg/m^3}$.

- The volume of an irregular object is measured by displacement of water, $V = V_2 - V_1$. The mass of a container must be subtracted, or it shows up as the constant term m_0 of the line.

Examples

Once you have studied the chapter's theory, move on to the guided examples. In the first you follow a worked solution step by step; in the next ones you make the choices yourself, and the hints help if you get stuck.

Example 1: Mass of an iron beam

A solid iron beam is $1.20\ \mathrm{m}$ long, and its cross-section is a square with a side of $4.0\ \mathrm{cm}$. What is the mass of the beam?

STEP 1/7

Given and find. The length is $L = 1.20\ \mathrm{m}$ and the side of the cross-section is $a = 4.0\ \mathrm{cm}$. From the table, the density of iron is $\rho = 7870\ \mathrm{kg/m^3}$. We are asked for the mass m of the beam.

STEP 2/7

Principle. The beam is iron all the way through, so its mass is directly proportional to volume, and the constant of proportionality is the density of iron. The volume is obtained as the product of the area of the cross-section, a^2, and the length.

STEP 3/7

Rearranged formula.

$$V = a^2 L$$
$$m = \rho V = \rho a^2 L$$

STEP 4/7

Unit conversions. The density is in $\mathrm{kg/m^3}$, so the lengths are converted to metres:

$$\begin{aligned} a &= 4.0\,\mathrm{cm} \\ &= 4.0 \cdot 0.01\,\mathrm{m} \\ &= 0.040\,\mathrm{m} \end{aligned}$$

The length is already in metres, and the mass comes out directly in kilograms.

STEP 5/7

Substitution.

$$\begin{aligned} m &= \rho a^2 L \\ &= 7870\,\tfrac{\mathrm{kg}}{\mathrm{m^3}} \cdot (0.040\,\mathrm{m})^2 \cdot 1.20\,\mathrm{m} \\ &= 7870\,\tfrac{\mathrm{kg}}{\mathrm{m^3}} \cdot 1.92 \cdot 10^{-3}\,\mathrm{m^3} \\ &\approx 15.11\,\mathrm{kg} \end{aligned}$$

STEP 6/7

Result. The mass of the beam is about $15\ \mathrm{kg}$. The result is given with two significant figures, because the side $4.0\ \mathrm{cm}$ is known to two significant figures; the length $1.20\ \mathrm{m}$ and the table density $7870\ \mathrm{kg/m^3}$ are known more precisely.

STEP 7/7

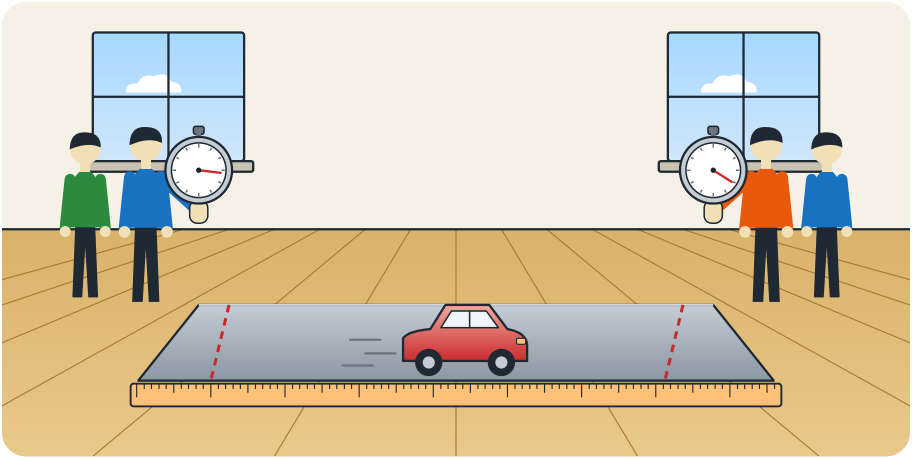
Reasonableness. The volume of the beam is $1.92\ \mathrm{dm^3}$, about two litres, and the same volume of water would have a mass of about $1.9\ \mathrm{kg}$. Iron is about 7.9 times as dense as water, so the estimate is $7.9 \cdot 1.9\ \mathrm{kg} \approx 15\ \mathrm{kg}$. The result matches.

FY1 · CHAPTER 8

Your own experimental investigation: planning, carrying out and reporting

> **Why** do two groups get different speeds for the same toy car, even though both measure carefully?

Simulation, tasks and model solutions online:
`ruukkilab.fi/en/fy1-8.html`



IN THIS LESSON, YOU WILL LEARN TO

- formulate a research question and a testable hypothesis.
- plan a fair experiment: choose the variables, the measuring range and the repeats.
- record measurements in a lab record and report a result with its uncertainty.

PRIOR KNOWLEDGE

- A measurement result has an uncertainty, and repeating the measurement gives a mean
  chapter *Measuring and the reliability of a measurement*
- A straight line is fitted to the data points, and its slope tells the rate of change
  chapter *Graphing and fitting a straight line*

Core path: follow the steps in order. Besides calculations, the core path includes investigations, explanations and graphs. Tasks marked *Optional* give extra practice. Optional tasks: 5.

Think about it

Two groups measure the speed of the same toy car with a hand stopwatch. Group A runs the car once over a distance of $0.50\ \mathrm{m}$, gets a time of $1.0\ \mathrm{s}$ and calculates a speed of $0.50\ \mathrm{m/s}$. Group B runs five different distances,

three times each. Over $2.00\ \mathrm{m}$ the mean time is $4.73\ \mathrm{s}$, and group B calculates a speed of $0.42\ \mathrm{m/s}$. The car and the batteries are the same.

> **Why do the results differ, and how should an investigation be designed so that its result can be trusted?**

Theory

Key idea

Physics is an experimental science. Physical claims are judged by observations and measurements. The methods of a good investigation are described so precisely that others can test the results and, where possible, repeat the measurements. An **investigation** is a systematic way of answering a question by measuring. It begins with a question and ends with a conclusion that the measurement results support. A good investigation is not one where the assumption turns out to be right, but one where the measurement could have shown the assumption to be wrong.



Researchers examine the laser setup of the LIGO gravitational-wave detector in a clean room at Hanford, USA, in 2019.

Photo: Nutsinee Kijbunchoo, CC BY 4.0

Research question and hypothesis

A research question is a question that a measurement can answer: "How does the travel time of a toy car depend on the length of the run?" The question "Which car is best?" is no good, because it cannot be measured.

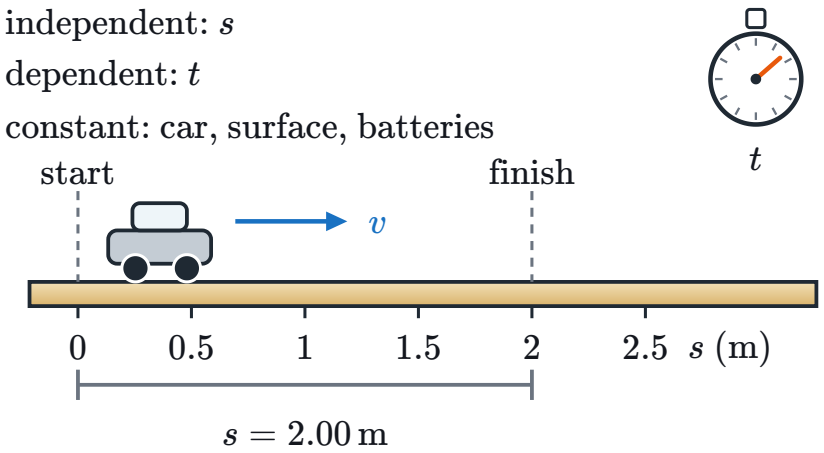
A is a reasoned, testable prediction that answers the research question. For example: "The car moves along the track at constant speed, so the distance it travels is directly proportional to time, $s = vt$." A hypothesis must lead to something measurable that could turn out to be wrong. The prediction "the car goes somehow" is not a hypothesis, because no measurement can disprove it.

Variables

A **variable** is a quantity or other factor that can vary in an investigation. In an experiment you change one variable, measure another and keep everything else the same:

- **independent variable**: the experimenter chooses its values, here the distance s
- **dependent variable**: its value is read from an instrument, here the time t
- **controlled variables**: stay the same throughout the experiment, here the car, the batteries, the surface and the way of starting.

If two variables change at the same time, for example the distance and the batteries, you cannot tell what caused the change in the result. That is why an experiment is fair only when just the independent variable changes.



Planning before measuring

Background information on the phenomenon and the methods is gathered before planning, with source criticism: judge who produced the information and why, how it is justified, whether it is up to date and whether other sources support it. Before measuring, the plan answers these questions: what is measured and with which instrument, which values are chosen for the independent variable and how many, how many repeats are made for each value, and what is kept constant. Try the plan

out first with a trial run. It tells you whether the measurement is possible at all, and it gives a rough estimate of the result.

Choose the measuring range so that the quantity being measured is clearly larger than the uncertainty of the measurement (see the section *Relative uncertainty and choosing the measuring range*). Choose several values at equal intervals, because one or two points do not show whether the dependence is a straight line.

Lab record

A **lab record** is a log made during the experiment, not from memory afterwards. It contains the research question, the date, the equipment and its accuracy, the controlled variables, the measurements as a table with units, and observations, for example "the car veered sideways".

Record the readings as they are, so that the record stays objective and is not shaped by your own expectations. Do not quietly delete or correct a measurement that looks odd: mark it and write down what happened. A short example:

s (m)	t_1 (s)	t_2 (s)	t_3 (s)	note
0.50	1.3	1.1	1.4	hard to time
1.00	2.5	2.3	2.6	
1.50	3.7	3.5	3.6	

Report

A describes the investigation so that someone else can repeat it and judge the conclusion. Its parts are:

1. research question and hypothesis
2. method: equipment, variables and procedure

3. results: table, graph and the value of the quantity with its uncertainty
4. discussion: sources of error and what you would do differently next time
5. conclusion: the answer to the research question and whether the results support the hypothesis.

A conclusion is only as strong as the data. If the measurements agree with the prediction within the uncertainty, the hypothesis has gained support, but it has not been proved.

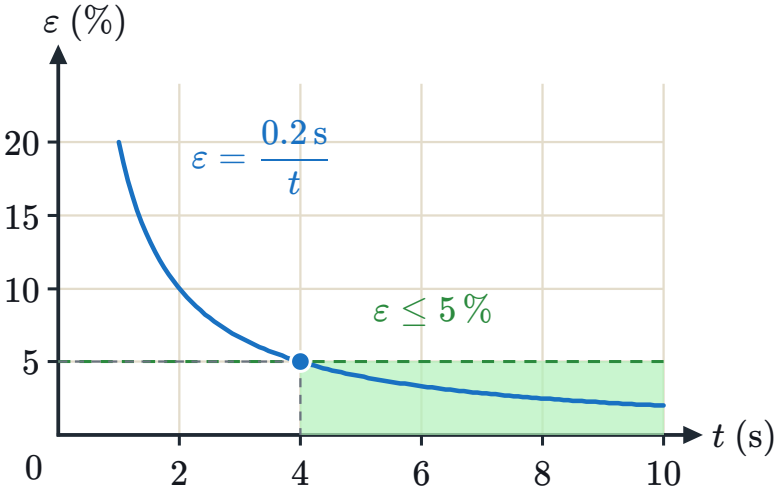
Relative uncertainty and choosing the measuring range

The uncertainty Δt of a time measured by hand comes mainly from reaction time, and it is almost the same in a short and a long measurement. The quality of a measurement is therefore described by the **relative uncertainty**, the uncertainty compared with the measured value. To keep it at most $\varepsilon_{\max}$, the measured time must be at least

$$\varepsilon = \frac{\Delta t}{t}$$

$$t \geq \frac{\Delta t}{\varepsilon_{\max}}$$

where ε is the relative uncertainty (often given as a percentage), Δt the uncertainty of the time (s) and t the measured time (s). With an uncertainty $\Delta t = 0.2\,\mathrm{s}$ and a limit of $5\,\%$, the time must be at least $4\,\mathrm{s}$.



The required distance follows from the model of uniform motion $s = vt$. If the measurement is made at several distances, a straight line is fitted to the data points (chapter *Graphing and fitting a straight line*). When t is on the horizontal axis and s on the vertical axis, the slope of the line is the speed v. The shortest valid distance is the one at which the relative uncertainty just stays within the chosen limit $\varepsilon_{\max}$. On a shorter distance the relative uncertainty exceeds this limit, and the result is too imprecise.

Common misconception

Many people think an investigation has failed if the measurement does not support the hypothesis. This is not so: if a measurement disproves the hypothesis, the investigator knows something they did not know before. A failed investigation is one from which nothing can be concluded, because the variables were not controlled or the measurements are too uncertain.

Another common mistake is to treat one measurement as enough. A single reading does not show how much the result varies, so the spread caused by random error cannot be estimated. A third mistake is to pick

the measurements that fit the result and leave out the others: all measurements are recorded, and an outlier is explained or marked.

Summary

- An investigation begins with a question that can be measured and a testable hypothesis that could turn out to be wrong.
- In a fair experiment you change one variable, measure another and keep the others constant.
- The measurement is planned in advance: measuring range, number of points, repeats and controlled variables. A trial run checks the plan.
- The lab record contains the readings, units, equipment and observations, written during the experiment. No measurement is quietly deleted.
- The relative uncertainty $\varepsilon = \Delta t : t$ gets smaller as the measured time gets longer; therefore the time must be at least $\Delta t : \varepsilon_{\max}$.
- A report gives the question, method, results, discussion and conclusion so that someone else can repeat the investigation.

Examples

Once you have studied the chapter's theory, move on to the guided examples. In the first you follow a worked solution step by step; in the next ones you make the choices yourself, and the hints help if you get stuck.

Example 1: Investigating a toy car's speed

Students investigate the motion of a toy car. In a trial run the speed of the car is about $0.42\,\mathrm{m/s}$. The time is measured with a hand stopwatch, and the timing uncertainty is $\Delta t = 0.2\,\mathrm{s}$. The relative uncertainty of the time measurement should be at most $5.0\,\%$. What is the shortest distance the car must travel?

STEP 1/7

Given and unknown. Speed $v = 0.42\,\mathrm{m/s}$, uncertainty of the time $\Delta t = 0.2\,\mathrm{s}$ and largest allowed relative uncertainty $\varepsilon_{\max} = 5.0\,\% = 0.050$. We are asked for the shortest distance $s_{\min}$.

STEP 2/7

Principle. The relative uncertainty gets smaller as the measured time gets longer, because Δt stays the same. First calculate the shortest allowed time from the uncertainty limit. The car moves at constant speed, so the model of uniform motion $s = vt$ gives the distance from the time.

STEP 3/7

Rearranged formula.

$$t_{\min} = \frac{\Delta t}{\varepsilon_{\max}}$$

$$s_{\min} = v\,t_{\min} = \frac{v\,\Delta t}{\varepsilon_{\max}}$$

STEP 4/7

Unit conversions. Not needed: the speed is in $\mathrm{m/s}$ and the time in seconds, so the distance comes out in metres. The percentage has been converted to a decimal number: $5.0\,\% = 0.050$.

STEP 5/7

Substitution.

$$\begin{aligned} t_{\min} &= \frac{\Delta t}{\varepsilon_{\max}} \\ &= \frac{0.2\,\mathrm{s}}{0.050} = 4\,\mathrm{s} \\ s_{\min} &= v\, t_{\min} \\ &= 0.42\,\tfrac{\mathrm{m}}{\mathrm{s}} \cdot 4\,\mathrm{s} \\ &= 1.68\,\mathrm{m} \end{aligned}$$

STEP 6/7

Result. The distance must be at least $1.7\,\mathrm{m}$. This is a design estimate whose precision is limited by the one-figure uncertainty $0.2\,\mathrm{s}$. The minimum distance of a design estimate is therefore given to practical precision, here two figures like the speed $0.42\,\mathrm{m/s}$, and rounded up so that the limit is surely met.

STEP 7/7

Reasonableness. Over $1.7\,\mathrm{m}$ the travel time of the car is about $1.7\,\mathrm{m} : 0.42\,\mathrm{m/s} \approx 4.05\,\mathrm{s}$, so

$$\begin{aligned} \varepsilon &= 0.2\,\mathrm{s} : 4.05\,\mathrm{s} \\ &\approx 4.9\,\% \end{aligned}$$

which is just within the limit. Over $0.50\,\mathrm{m}$ the time would be only $1.2\,\mathrm{s}$ and the uncertainty about $17\,\%$, which is why the short distance was a poor choice.

Theory and worked examples from Ruukkilabra

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